

# SS1 GEOGRAPHY LESSON NOTES

## FIRST TERM

### WEEK 1: INTRODUCTION TO GEOGRAPHY: MEANING, SCOPE, AND VALUE

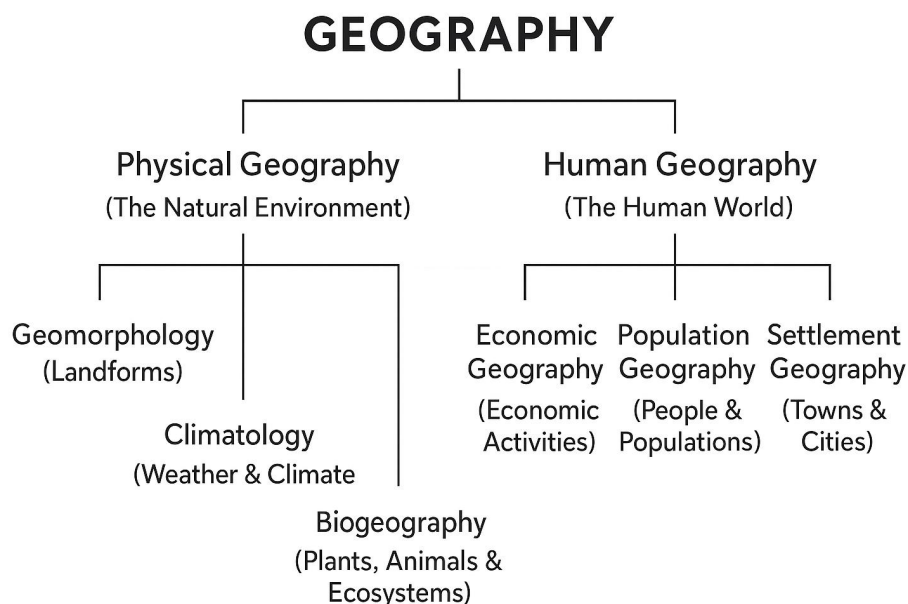
#### 1.0 MEANING OF GEOGRAPHY

Geography is derived from the Greek words 'Geo' (Earth) and 'graphia' (writing or description). It is the **integrated science** that studies the Earth's landscapes, peoples, places, and environments. It goes beyond mere description to explain **how human and physical phenomena are distributed over the Earth's surface and how they interact.**

- **Core Concept:** Geography answers two fundamental questions:
  - a. **Where** are things located? (Spatial Distribution)
  - b. **Why** are they there? (Spatial Processes and Relationships)

#### 2.0 THE SCOPE OF GEOGRAPHY: THE TWO MAIN BRANCHES

The scope of geography is vast, bridging the natural sciences, social sciences, and humanities. It is primarily divided into two interconnected branches:



**2.1 Physical Geography:** Focuses on the natural environment and the processes that shape it.

- **Geomorphology:** The study of landforms (e.g., mountains, valleys, rivers) and the processes that create them (e.g., erosion, volcanism).
- **Climatology:** The study of long-term weather patterns (climate) and atmospheric conditions.
- **Biogeography:** The study of the distribution of plants and animals.
- **Hydrology:** The study of water on Earth (rivers, lakes, oceans, groundwater).
- **Pedology:** The study of soils.

**2.2 Human Geography:** Examines human activities and their relationship with the physical environment.

- **Economic Geography:** Studies economic activities (agriculture, mining, industry, trade).
- **Population Geography:** Analyzes population distribution, density, growth, and migration.
- **Settlement Geography:** Examines the patterns and functions of villages, towns, and cities.
- **Political Geography:** Studies governments, territories, and boundaries.

### ***3.0 THE VALUE AND IMPORTANCE OF GEOGRAPHY***

Geography is not just about memorizing places on a map; it is a vital tool for understanding and solving real-world problems.

1. **Environmental Understanding and Management:** It enables us to understand natural hazards (floods, earthquakes), manage resources sustainably, and address environmental issues like climate change and pollution.
2. **Spatial Awareness and Global Citizenship:** In an interconnected world, geography fosters an understanding of different cultures, economies, and global issues, promoting tolerance and informed citizenship.
3. **Critical Career Applications:** Geographers work in diverse fields:
  - **Urban and Regional Planning:** Designing sustainable cities.

- **Geographic Information Systems (GIS):** Mapping and analyzing spatial data for businesses and governments.
  - **Environmental Consultancy:** Assessing environmental impact.
  - **Disaster Management:** Planning for and responding to natural disasters.
  - **Logistics and Transportation:** Optimizing supply chains.
4. **Interdisciplinary Bridge:** Geography provides essential context for other subjects. History is understood through its geographical setting; Economics depends on the location of resources and markets.
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### ***EVALUATION (WEEK 1)***

1. Define Geography, explaining the meaning of its Greek roots.
2. Differentiate, with three examples each, between Physical Geography and Human Geography.
3. Explain three reasons why Geography is a valuable subject for a secondary school student in Nigeria.
4. A geographer studying the pattern of commercial banks in Lagos is specializing in which branch of Geography? Explain your answer.

### ***ASSIGNMENT (WEEK 1)***

1. Create a detailed poster or diagram illustrating the main branches and sub-branches of Geography.
  2. Write a short essay (250 words) on the topic: "How can the study of Geography help in solving the problem of flooding in my local community?"
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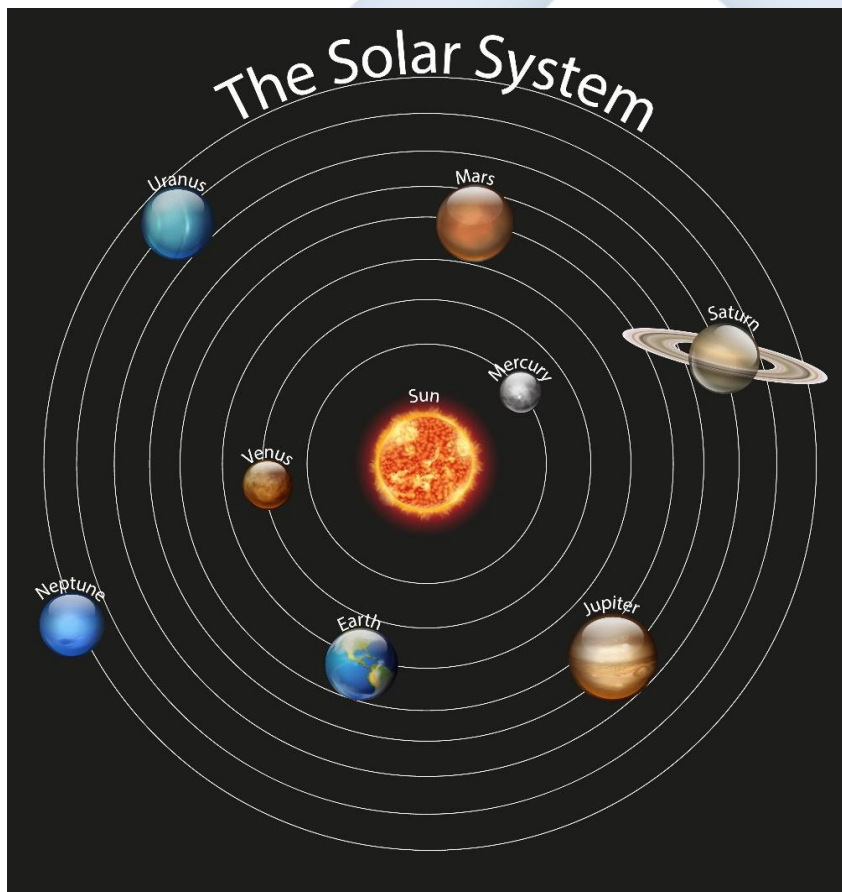
## WEEK 2: THE EARTH AND OTHER PLANETS

### 1.0 THE SOLAR SYSTEM: OUR COSMIC NEIGHBORHOOD

The Solar System is a gravitationally bound system comprising the **Sun** and the objects orbiting it. These include:

- **Planets:** Eight large celestial bodies.
- **Dwarf Planets:** e.g., Pluto.
- **Moons (Satellites):** Natural satellites orbiting planets.
- **Asteroids & Meteoroids:** Rocky and metallic objects.
- **Comets:** Icy bodies that develop glowing tails near the Sun.

**Illustration: The Solar System (A labeled diagram showing the Sun and planets in order)**



## 2.0 CLASSIFICATION AND FEATURES OF PLANETS

Planets are classified based on their position, composition, and size.

### 2.1 Terrestrial (Inner/Rocky) Planets: Mercury, Venus, Earth, Mars.

- **Characteristics:** Closer to the Sun, smaller, denser, rocky surfaces, thin atmospheres (except Venus), few or no moons.

### 2.2 Jovian (Outer/Gas Giant) Planets: Jupiter, Saturn, Uranus, Neptune.

- **Characteristics:** Farther from the Sun, very large, low density, composed mostly of gases (hydrogen, helium), have ring systems, and numerous moons.

#### Comprehensive Table of Planetary Features:

| Planet         | Type        | Diameter<br>(Earth=1) | Key Features &<br>Atmosphere                                       | Moons | Special Notes                          |
|----------------|-------------|-----------------------|--------------------------------------------------------------------|-------|----------------------------------------|
| <b>Mercury</b> | Terrestrial | 0.38                  | Rocky, no atmosphere, extreme temperature swings                   | 0     | Closest to the Sun; heavily cratered   |
| <b>Venus</b>   | Terrestrial | 0.95                  | Thick, toxic CO <sub>2</sub> atmosphere; runaway greenhouse effect | 0     | Hottest planet; "Earth's twin" in size |
| <b>Earth</b>   | Terrestrial | 1.00                  | Nitrogen-Oxygen atmosphere; liquid water; life                     | 1     | Only known planet with life            |
| <b>Mars</b>    | Terrestrial | 0.53                  | Thin CO <sub>2</sub> atmosphere; reddish from iron oxide (rust)    | 2     | "Red Planet"; has polar ice caps       |
| <b>Jupiter</b> | Jovian      | 11.21                 | Giant gas planet; Great Red Spot (a giant storm)                   | 95+   | Largest planet; has faint rings        |
| <b>Saturn</b>  | Jovian      | 9.45                  | Gaseous; spectacular                                               | 83+   | Least dense planet                     |

|                |        |      |                                                |    |                                            |
|----------------|--------|------|------------------------------------------------|----|--------------------------------------------|
|                |        |      | ring system of ice and rock                    |    | (would float in water)                     |
| <b>Uranus</b>  | Jovian | 4.01 | Icy atmosphere (methane); rotates on its side  | 27 | Appears blue-green due to methane          |
| <b>Neptune</b> | Jovian | 3.88 | Icy/gaseous; fastest winds in the solar system | 14 | Discovered through mathematical prediction |

### ***3.0 EARTH'S UNIQUE POSITION: THE GOLDILOCKS ZONE***

Earth is uniquely suited for life because it is located in the "**Habitable Zone**" or "**Goldilocks Zone**" of our solar system.

- **Not too hot, not too cold:** It is at the right distance from the Sun for water to exist in its liquid state.
- **Protective Atmosphere:** The atmosphere provides breathable air, regulates temperature, and shields us from harmful solar radiation and meteoroids.
- **Magnetic Field:** Earth's magnetic field deflects most of the charged particles from the solar wind.

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### ***EVALUATION (WEEK 2)***

1. What is the Solar System? List the eight planets in order of their distance from the Sun.
2. State four key differences between Terrestrial and Jovian planets.
3. Explain why Earth is the only planet in our solar system known to support life, mentioning at least three reasons.
4. Which planet is known for its spectacular ring system, and which planet has the highest number of moons?

### ***ASSIGNMENT (WEEK 2)***

1. Draw a well-labeled diagram of the solar system. Use different colors to distinguish between the Terrestrial and Jovian planets.

2. Research and write a one-page report on the "Great Red Spot" of Jupiter, explaining what it is and how long it has been observed.
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## WEEK 3: EARTH'S ROTATION AND REVOLUTION

### 1.0 DEFINITIONS AND KEY DIFFERENCES

The Earth has two primary motions that are fundamental to our experience of time and seasons.

- **Rotation:** The **spinning** of the Earth on its imaginary **axis** (an line through the North and South Poles). It takes approximately **24 hours** to complete one rotation.
- **Revolution:** The **movement** of the Earth around the **Sun** in a slightly elliptical path called an **orbit**. It takes **365 ¼ days** (one year) to complete one revolution.

#### Comparison Table:

| Criteria               | Rotation                  | Revolution                             |
|------------------------|---------------------------|----------------------------------------|
| <b>Definition</b>      | Spinning on its axis      | Movement around the Sun                |
| <b>Time Taken</b>      | 24 hours (1 day)          | 365 ¼ days (1 year)                    |
| <b>Path</b>            | Around the internal axis  | Along an elliptical orbit              |
| <b>Primary Effects</b> | Day and Night; Time Zones | Seasons; Varying Length of Day & Night |

### 2.0 EFFECTS OF ROTATION

1. **Day and Night:** As the Earth rotates, only the half facing the Sun experiences day, while the other half experiences night. This creates the daily cycle of daylight and darkness.
2. **Apparent Movement of the Sun:** The Sun appears to rise in the east and set in the west due to the Earth's west-to-east rotation.
3. **Time Zones:** The Earth is divided into 24 time zones, each roughly 15° of longitude apart ( $360^\circ / 24 \text{ hours} = 15^\circ/\text{hour}$ ). This means that as you move east, time advances, and as you move west, it goes back.
4. **The Coriolis Effect:** This is the deflection of moving objects (like winds and ocean currents) caused by the Earth's rotation.
  - **Deflection:** To the **right** in the **Northern Hemisphere**.
  - **Deflection:** To the **left** in the **Southern Hemisphere**.

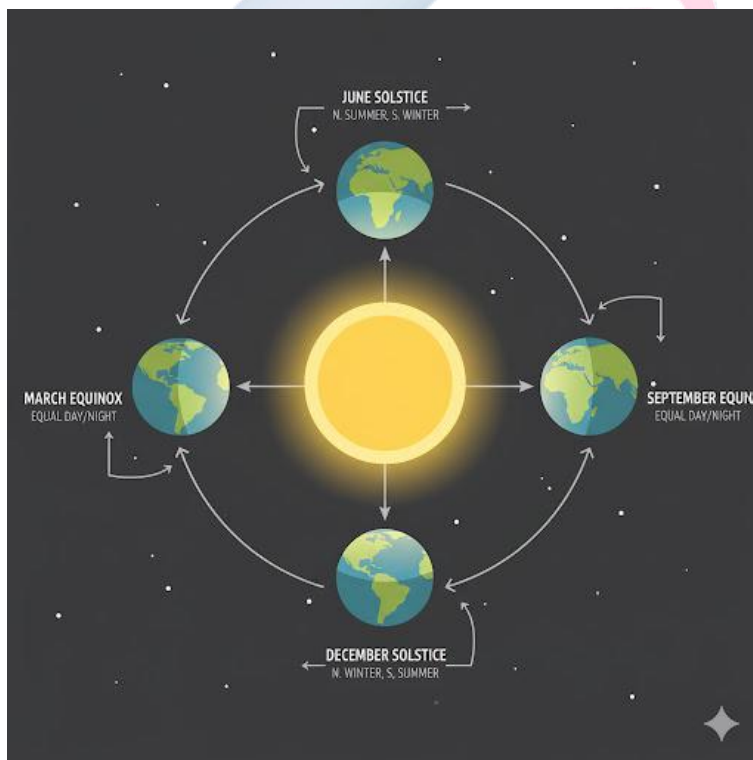


### 3.0 EFFECTS OF REVOLUTION

The Earth's **tilted axis (23.5°)** combined with its revolution causes the seasons.

1. **Seasons:** The hemisphere tilted *towards* the Sun experiences summer (more direct sunlight, longer days). The hemisphere tilted *away* experiences winter.
2. **Solstices and Equinoxes:**
  - **Summer Solstice (June 21):** The North Pole is tilted most towards the Sun. Longest day in the Northern Hemisphere.
  - **Winter Solstice (December 22):** The North Pole is tilted most away from the Sun. Shortest day in the Northern Hemisphere.
  - **Equinoxes (March 21 & September 23):** The Sun is directly above the Equator. Day and night are of nearly equal length all over the world.

**Illustration: Earth's Revolution and Seasons** *(A diagram showing Earth's position at the four key points in its orbit, with the axis tilt clearly visible, explaining the season in each hemisphere.)*



3. **Leap Year:** The extra quarter-day each year is accounted for by adding an extra day (February 29) every four years.

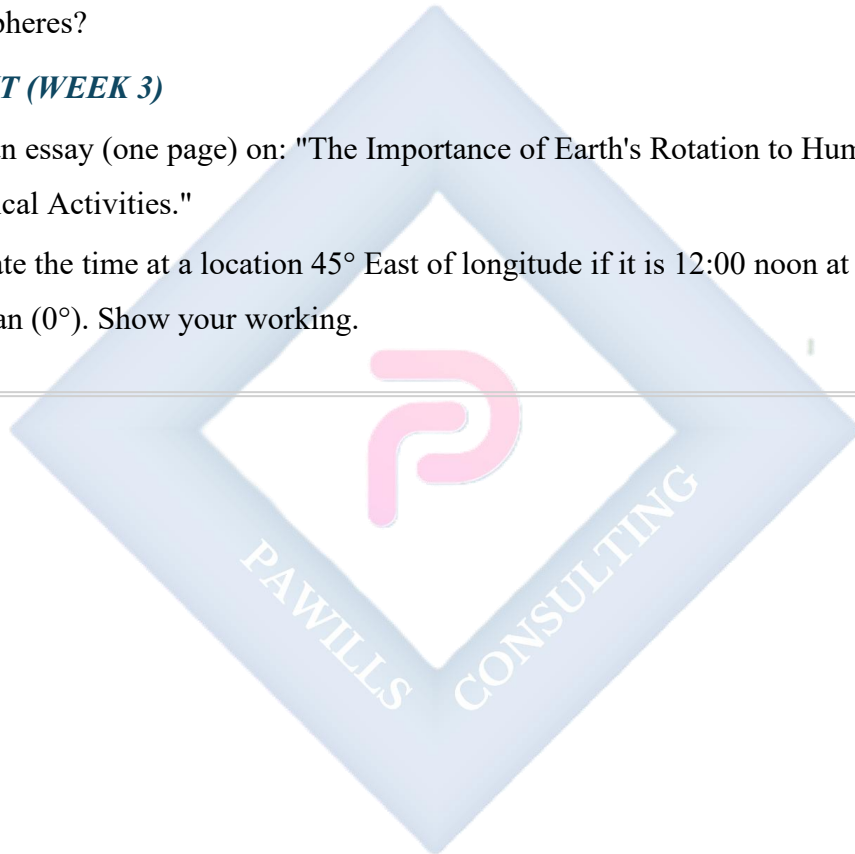
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### ***EVALUATION (WEEK 3)***

1. Differentiate clearly between the Earth's rotation and revolution.
2. List three major effects of the Earth's rotation.
3. Explain, with the aid of a diagram, how the revolution of the Earth and its tilted axis cause the seasons.
4. What is the Coriolis Effect, and how does it manifest in the Northern and Southern Hemispheres?

### ***ASSIGNMENT (WEEK 3)***

1. Write an essay (one page) on: "The Importance of Earth's Rotation to Human and Biological Activities."
2. Calculate the time at a location  $45^\circ$  East of longitude if it is 12:00 noon at the Prime Meridian ( $0^\circ$ ). Show your working.



## WEEK 4: LATITUDES AND LONGITUDES

### 1.0 LATITUDE: MEASURING NORTH AND SOUTH

- **Definition:** Latitude is the angular distance, measured in degrees, north or south of the Equator.
- **Key Features:**
  - The **Equator** is the starting point at **0° Latitude**.
  - Lines of latitude are called **parallels** because they run parallel to each other and never meet.
  - They are complete circles, getting smaller towards the poles.
  - There are **90° North** (North Pole) and **90° South** (South Pole).

#### Major Lines of Latitude:

1. **Equator (0°)**
2. **Tropic of Cancer (23.5° N)**
3. **Tropic of Capricorn (23.5° S)**
4. **Arctic Circle (66.5° N)**
5. **Antarctic Circle (66.5° S)**
6. **North Pole (90° N)**
7. **South Pole (90° S)**

### 2.0 LONGITUDE: MEASURING EAST AND WEST

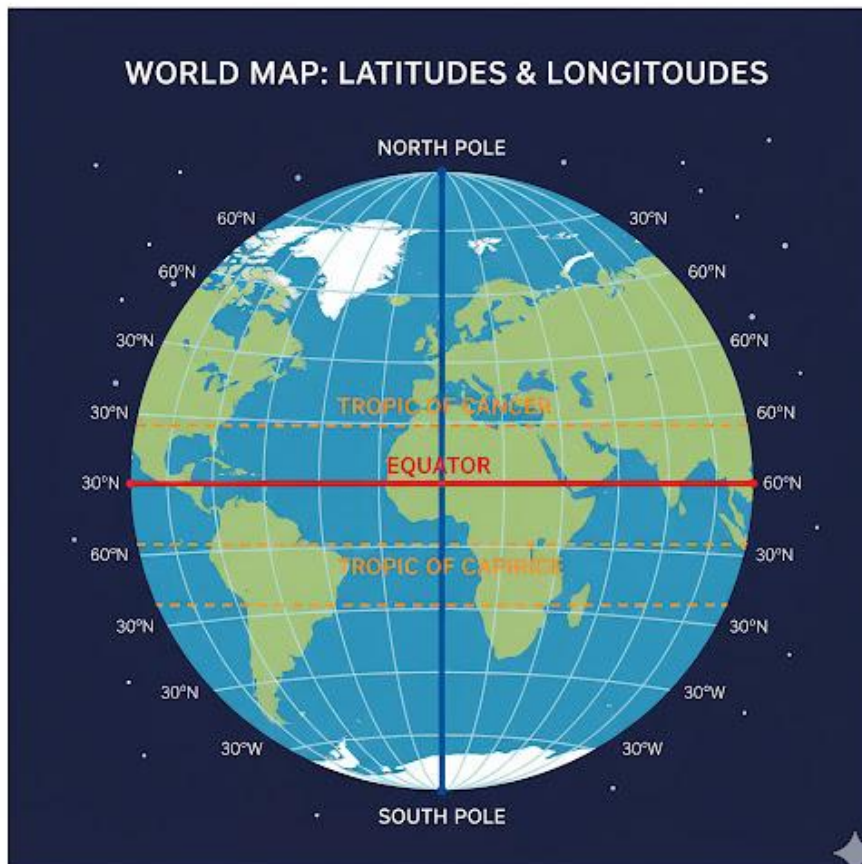
- **Definition:** Longitude is the angular distance, measured in degrees, east or west of the Prime Meridian.
- **Key Features:**
  - The **Prime Meridian** at **0° Longitude** passes through Greenwich, England.
  - Lines of longitude are called **meridians**. They are semi-circles that converge at the poles.
  - There are **180° East** and **180° West** of the Prime Meridian, meeting at the **International Date Line**.

### 3.0 LOCATING PLACES USING COORDINATES

Any location on Earth can be precisely identified using its **geographic coordinates** (Latitude, Longitude).

- **Format:** Always **Latitude first, then Longitude**.
- **Example:** Abuja, Nigeria: 9° N, 7° E.

**Illustration: World Map with Latitudes and Longitudes** (*A diagram showing a globe with a grid of parallels and meridians, highlighting the Equator, Prime Meridian, Tropics, and Poles.*)



### 4.0 IMPORTANCE OF LATITUDE AND LONGITUDE

1. **Pinpoint Navigation:** Essential for GPS, aviation, and maritime navigation.
2. **Time Calculation:** Each 15° of longitude represents a one-hour time difference. The Prime Meridian defines Greenwich Mean Time (GMT).

3. **Climate Classification:** Latitudes are used to define the world's major climate zones: **Tropical** (low latitudes), **Temperate** (mid-latitudes), and **Polar** (high latitudes).
  4. **Map Making (Cartography):** The fundamental grid for creating accurate maps.
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#### ***EVALUATION (WEEK 4)***

1. Define latitude and longitude.
2. State four differences between lines of latitude and lines of longitude.
3. Name the five major lines of latitude and state their angular distances.
4. If the time at 0° longitude (Greenwich) is 10:00 am, what will be the time at 45° E longitude? Show your calculations.

#### ***ASSIGNMENT (WEEK 4)***

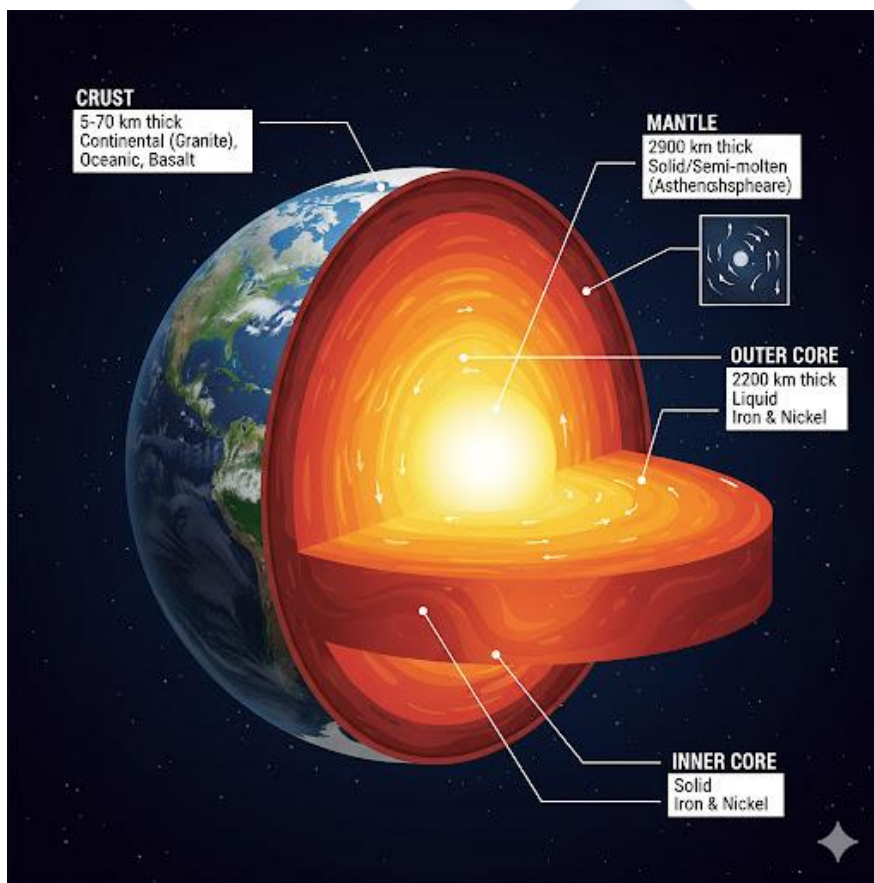
1. Draw an outline map of the world. On it, clearly label and mark:
    - The Equator
    - Tropic of Cancer and Tropic of Capricorn
    - Arctic and Antarctic Circles
    - Prime Meridian and International Date Line
  2. Using an atlas, find and write down the approximate coordinates (latitude and longitude) of your hometown.
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## WEEK 6: THE EARTH'S STRUCTURE

### 1.0 THE CONCENTRIC LAYERS OF THE EARTH

The Earth is not a uniform sphere but is composed of a series of concentric layers, each with distinct chemical and physical properties.

**Illustration: The Internal Structure of the Earth** *(A detailed cross-section diagram showing the Crust, Mantle, Outer Core, and Inner Core, with labels for thickness, state of matter, and composition.)*



### 2.0 DETAILED DESCRIPTION OF THE LAYERS

#### 2.1 The Crust

- **Description:** The thin, solid, outermost layer.
- **Thickness:** 5-70 km (thinnest under oceans, thickest under mountains).
- **Composition:**

- **Continental Crust (SIAL):** Composed mainly of **Silica** and **Aluminium**. Less dense, mostly granite.
- **Oceanic Crust (SIMA):** Composed mainly of **Silica** and **Magnesium**. More dense, mostly basalt.

## 2.2 The Mantle

- **Description:** A thick layer of hot, solid rock that behaves plastically (like soft plastic) over geological time.
- **Thickness:** About 2,900 km.
- **Key Zone:** The **Asthenosphere** (upper part of the mantle) is partially molten and is the layer upon which the rigid crustal plates move.
- **Process: Convection Currents** within the mantle are the driving force for plate tectonics.

## 2.3 The Core

- **Description:** The innermost and hottest part of the Earth.
- **Composition:** Mostly **Iron** and **Nickel**.
- **Parts:**
  - **Outer Core:** Liquid. The movement of this liquid metal generates Earth's **magnetic field**.
  - **Inner Core:** Solid, due to immense pressure, despite temperatures reaching ~5,500°C.

### Summary Table of Earth's Layers:

| Layer         | State               | Thickness | Composition                        | Key Fact                      |
|---------------|---------------------|-----------|------------------------------------|-------------------------------|
| <b>Crust</b>  | Solid               | 5-70 km   | SIAL (Continental), SIMA (Oceanic) | Thinnest layer; we live on it |
| <b>Mantle</b> | Solid (but plastic) | ~2,900 km | Silicates of Iron & Magnesium      | Source of convection currents |
| <b>Outer</b>  | Liquid              | ~2,200    | Iron & Nickel                      | Generates Earth's             |



| Layer             | State | Thickness | Composition   | Key Fact                       |
|-------------------|-------|-----------|---------------|--------------------------------|
| <b>Core</b>       |       | km        |               | magnetic field                 |
| <b>Inner Core</b> | Solid | ~1,200 km | Iron & Nickel | Hottest part; immense pressure |

### ***3.0 THE EARTH'S ATMOSPHERE***

The atmosphere is a layer of gases held by Earth's gravity, crucial for life.

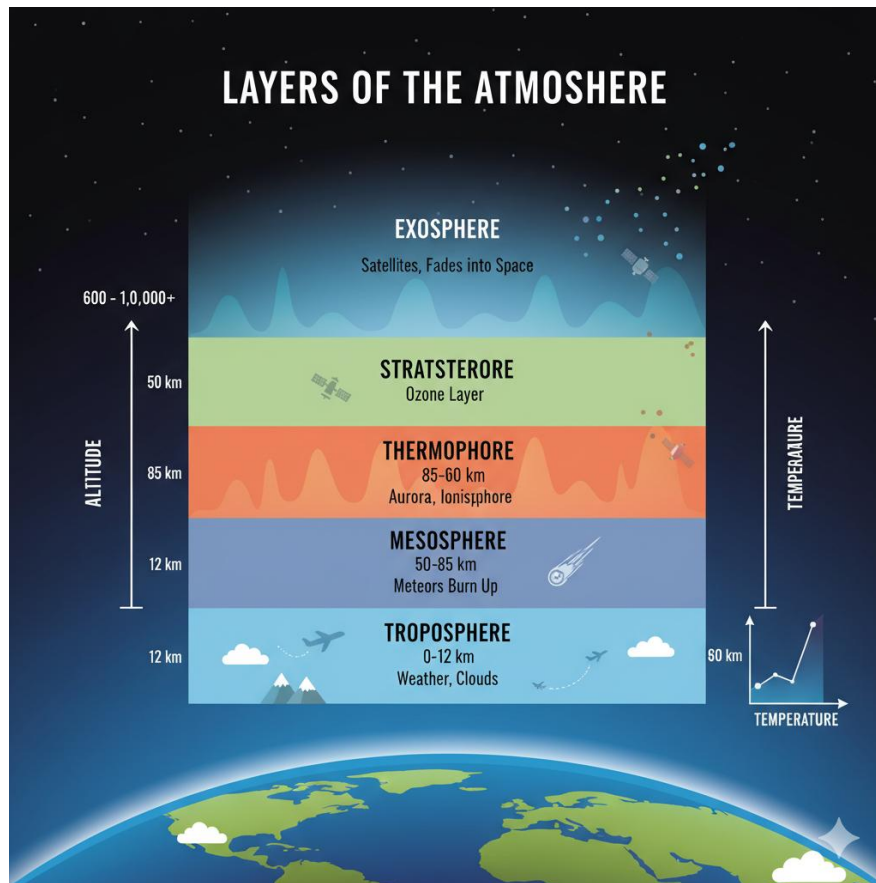
- **Composition:** 78% Nitrogen, 21% Oxygen, 1% other gases (Argon, CO<sub>2</sub>, etc.).

#### **Layers of the Atmosphere:**

1. **Troposphere:** Where we live; all weather occurs here.
2. **Stratosphere:** Contains the Ozone Layer, which absorbs UV radiation.
3. **Mesosphere:** Meteors burn up here.
4. **Thermosphere:** Northern Lights occur here; very high temperatures.
5. **Exosphere:** The outermost layer, merging into space.



**Illustration: The Layers of the Atmosphere** (A diagram showing the different layers stacked on top of each other, with labels for altitude and key characteristics.)



### **EVALUATION (WEEK 6)**

1. Name the three main concentric layers of the Earth and describe their physical state.
2. Differentiate between the Continental Crust and the Oceanic Crust in terms of composition and density.
3. What is the asthenosphere, and why is it important for plate tectonics?
4. Explain how the Earth's magnetic field is generated.

### **ASSIGNMENT (WEEK 6)**

1. Draw and label a detailed diagram of the Earth's internal structure.
2. Research and write a short paragraph on the "Ozone Layer," explaining its location and importance.

## WEEK 8: ROCKS

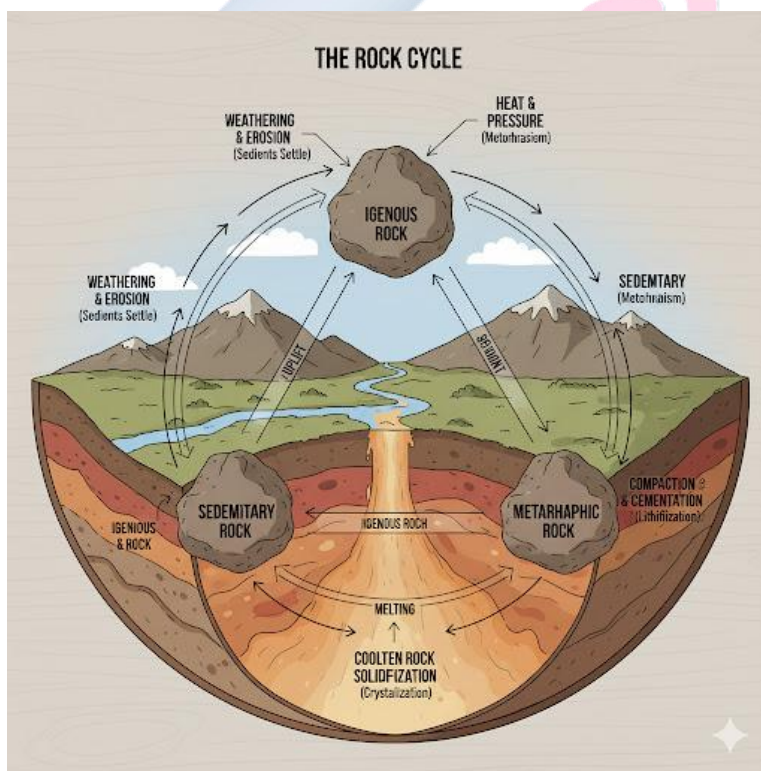
### 1.0 MEANING AND SIGNIFICANCE OF ROCKS

Rocks are naturally occurring solid aggregates of one or more minerals or mineraloids. They form the fundamental building blocks of the Earth's crust and provide a record of Earth's geological history. The study of rocks (Petrology) is crucial for understanding landscape formation, locating resources, and reconstructing past environments.

### 2.0 THE THREE PRIMARY ROCK TYPES: A GENETIC CLASSIFICATION

Rocks are classified into three major groups based on their origin and formation process.

**Illustration: The Rock Cycle** (A comprehensive diagram showing the interconnected processes that form and transform the three rock types, including melting, cooling, weathering/erosion, deposition, compaction/cementation, and heat/pressure.)



## *2.1 Igneous Rocks: The Primary Rocks*

- **Origin:** Formed from the cooling and solidification of molten rock material.
  - **Magma:** Molten rock beneath the Earth's surface.
  - **Lava:** Molten rock that reaches the Earth's surface.
- **Classification:**
  - **Intrusive (Plutonic) Igneous Rocks:**
    - **Formation:** Magma cools slowly deep within the Earth's crust.
    - **Texture:** Coarse-grained (large, visible crystals) due to slow cooling.
    - **Examples:** **Granite** (abundant in continental crust), **Gabbro** (abundant in oceanic crust), Diorite.
  - **Extrusive (Volcanic) Igneous Rocks:**
    - **Formation:** Lava cools rapidly on the Earth's surface.
    - **Texture:** Fine-grained, glassy, or vesicular (containing holes from gas bubbles).
    - **Examples:** **Basalt** (makes up most of the ocean floor), **Pumice** (so light it floats), **Obsidian** (volcanic glass).
- **Key Characteristics:**
  - Interlocking crystalline structure.
  - Never contain fossils.
  - Often form rugged, resistant landscapes.

## *2.2 Sedimentary Rocks: The Storytellers*

- **Origin:** Formed from the accumulation, compaction, and cementation of sediments.
- **Process:** **Weathering** → **Erosion** → **Transportation** → **Deposition** → **Lithification** (compaction & cementation).
- **Classification:**

- **Clastic (Detrital):** Formed from mechanical weathering debris.
  - **Examples: Sandstone** (sand-sized grains), **Shale** (clay-sized grains), **Conglomerate** (rounded gravel).
- **Chemical:** Formed by precipitation of minerals from water solution.
  - **Examples: Rock Salt** (halite), **Limestone** (travertine), **Gypsum**.
- **Organic (Biogenic):** Formed from the accumulation of plant or animal debris.
  - **Examples: Limestone** (from shell fragments), **Coal** (from compressed plant matter), Chalk.
- **Key Characteristics:**
  - Often **stratified** (arranged in layers or strata).
  - May contain **fossils**, providing a record of past life.
  - Often softer and more porous than igneous rocks.

### *2.3 Metamorphic Rocks: The Changed Rocks*

- **Origin:** Formed from the transformation of pre-existing rocks (igneous, sedimentary, or other metamorphic rocks) under the influence of high **heat**, high **pressure**, and/or chemically active fluids. This process, called **metamorphism**, occurs in the solid state (the rock does not melt).
- **Classification:**
  - **Foliated:** Minerals are aligned in parallel layers or bands due to directed pressure.
    - **Examples: Slate** (from Shale), **Schist**, **Gneiss** (from Granite).
  - **Non-Foliated:** No banding; minerals recrystallize without directed pressure.
    - **Examples: Marble** (from Limestone), **Quartzite** (from Sandstone).
- **Key Characteristics:**
  - Often harder and more compact than their parent rock.
  - May show foliation or a sparkling appearance from new mineral crystals.
  - Rarely contain fossils (usually destroyed by heat and pressure).

### ***3.0 THE ROCK CYCLE: A DYNAMIC SYSTEM***

The rock cycle is a continuous, never-ending process that illustrates how the three rock types are interrelated and how each can be transformed into another over geological time through various Earth processes.

### ***4.0 ECONOMIC AND CULTURAL IMPORTANCE OF ROCKS***

- **Construction:** Granite, limestone, and sandstone for buildings, monuments, and road aggregate.
  - **Energy Resources:** Coal for electricity; Uranium (from specific igneous and sedimentary rocks) for nuclear power.
  - **Industrial Minerals:**
    - Limestone is essential for making cement and steel.
    - Rock Salt is used for de-icing roads and in the chemical industry.
  - **Metallic Ores:** Rocks contain valuable metals like iron (Hematite), copper (Chalcopyrite), and aluminum (Bauxite).
  - **Gemstones and Decor:** Diamonds, rubies, marble, and slate for jewelry, sculpture, and countertops.
  - **Agriculture:** Limestone is crushed to neutralize acidic soils.
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### ***EVALUATION (WEEK 8)***

1. Define the term "rock" and explain why rocks are considered the record of Earth's history.
2. Describe how intrusive and extrusive igneous rocks are formed. Why do they have different textures?
3. With specific examples, differentiate between clastic, chemical, and organic sedimentary rocks.
4. Explain the process of metamorphism. How does it differ from the melting process that forms igneous rocks?
5. Why are sedimentary rocks the only type that commonly contain fossils?

### ***ASSIGNMENT (WEEK 8)***

1. Create a detailed, annotated diagram of the Rock Cycle. For each rock type, include at least two specific examples and briefly describe the process that forms it.
  2. Conduct a "Rock Hunt" around your school or home. Collect three different rock samples (if possible). For each, describe its physical properties (color, texture, hardness) and try to classify it as igneous, sedimentary, or metamorphic, justifying your reasoning.
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## WEEK 9: TOWN AND VILLAGE

### 1.0 DEFINITIONS AND CORE CONCEPTS

- **Village:** A **rural settlement** with a small population density, where the majority of inhabitants are engaged in **primary activities** (agriculture, fishing, forestry, mining). It is characterized by a close-knit social structure and limited infrastructure.
- **Town:** An **urban settlement** that is larger than a village but smaller than a city. It has a higher population density, diverse **economic base** (including secondary and tertiary activities), and more developed infrastructure. It often serves as an administrative or commercial center for surrounding villages.

### 2.0 CHARACTERISTICS: A COMPARATIVE ANALYSIS

#### Comprehensive Comparison Table:

| Feature               | Village (Rural)                                                         | Town (Urban)                                                                                         |
|-----------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Population</b>     | Small size, low density, often homogeneous.                             | Larger size, high density, often diverse/multi-ethnic.                                               |
| <b>Economy</b>        | <b>Primary Sector Dominant:</b><br>Farming, fishing, herding.           | <b>Secondary &amp; Tertiary Sectors Dominant:</b> Manufacturing, commerce, services, administration. |
| <b>Infrastructure</b> | Basic and limited: unpaved roads, wells/boreholes, limited electricity. | Developed: paved roads, pipe-borne water, stable electricity, sewage systems.                        |
| <b>Social Life</b>    | Strong community bonds, traditional values, slow pace of life.          | More impersonal, diverse cultures, modern values, fast-paced lifestyle.                              |
| <b>Housing</b>        | Simple, often using local materials (mud, thatch).                      | Complex, using modern materials (cement, glass, steel); varied architectural styles.                 |
| <b>Administration</b> | Traditional leadership (Chiefs,                                         | Formal government structures (Local                                                                  |



|                  |                                                         |                                                                        |
|------------------|---------------------------------------------------------|------------------------------------------------------------------------|
|                  | Elders).                                                | Government Council).                                                   |
| <b>Amenities</b> | Few schools (often primary only), basic health centers. | Multiple schools (primary, secondary), hospitals, banks, supermarkets. |

### ***3.0 FUNCTIONS AND INTERDEPENDENCE***

- **Functions of a Village:**
  - **Agricultural Production:** Supplies food and raw materials to urban areas.
  - **Cultural Preservation:** Maintains traditional customs, languages, and crafts.
  - **Resource Base:** Provides land, water, and mineral resources.
- **Functions of a Town:**
  - **Administrative Center:** Hosts local government offices and services.
  - **Commercial Hub:** Provides markets, banking, and retail services for a wider region.
  - **Industrial and Service Center:** Offers employment in factories, schools, and hospitals.
  - **Transport Node:** Often located at the intersection of roads or railways.
- **Interdependence:** Towns and villages exist in a **symbiotic relationship**.
  - **Villages → Towns:** Supply food, raw materials, and labor.
  - **Towns → Villages:** Provide manufactured goods, markets for village produce, higher education, specialized healthcare, and administrative services.

### ***4.0 PROBLEMS AND SUSTAINABLE SOLUTIONS***

- **Problems in Villages:**
  - a. **Rural-Urban Migration:** Exodus of youth to cities, leading to an aging population.
  - b. **Poverty and Unemployment:** Limited economic opportunities.
  - c. **Poor Infrastructure:** Lack of good roads, electricity, and internet connectivity.

- **Solutions for Villages:**

- **Rural Development Programs:** Investing in infrastructure (roads, electricity, water).
- **Agricultural Modernization:** Providing subsidies, training, and access to markets.
- **Establishing Rural Industries:** To create jobs and add value to local products.

- **Problems in Towns:**

- a. **Overcrowding and Congestion:** Leading to traffic jams and long commute times.
- b. **Inadequate Housing and Slums:** Resulting from rapid urbanization.
- c. **Environmental Pollution:** Air, water, and land pollution from industries, vehicles, and waste.
- d. **High Cost of Living and Social Vices:** Unemployment can lead to increased crime rates.

- **Solutions for Towns:**

- **Effective Urban Planning:** Zoning laws, development of satellite towns, and efficient public transport systems.
- **Provision of Affordable Housing:** Government-backed housing projects.
- **Strict Environmental Regulations:** And investment in waste management and recycling.

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### ***EVALUATION (WEEK 9)***

1. Differentiate between a town and a village based on population, economic activities, and infrastructure.
2. Explain the concept of "rural-urban interdependence" with two examples of flows from the village to the town and two from the town to the village.
3. Identify three major problems associated with rapid urban growth (urbanization) in towns.
4. Suggest two sustainable solutions to the problem of rural-urban migration.

### ***ASSIGNMENT (WEEK 9)***

1. Write a comparative profile (400 words) of a specific village and a specific town you know or have researched. Focus on their physical layout, economic base, social life, and the main challenges they face.
2. Design a plan for a "Model Village" that would discourage rural-urban migration. Your plan should include proposals for infrastructure, economic opportunities, and social amenities.



## WEEK 10: THE LOCAL GOVERNMENT AREA (LGA)

### 1.0 CONCEPT AND PURPOSE OF LOCAL GOVERNMENT

- **Definition:** A **Local Government Area (LGA)** is the third tier of government in Nigeria (below the Federal and State governments), established to bring governance and development closer to the people at the grassroots level.
- **Purpose (Rationale for Creation):**
  - **Decentralization of Power:** To avoid concentration of power at the federal and state levels.
  - **Grassroots Development:** To address the specific needs and priorities of local communities.
  - **Political Participation:** To encourage citizen involvement in governance.
  - **Effective Service Delivery:** To provide essential services efficiently at the local level.

### 2.0 STRUCTURE AND ADMINISTRATION OF AN LGA

The structure is designed to mirror the executive and legislative arms at the national and state levels.

- **The Executive Arm:**
  - **Chairman:** The Chief Executive and political head of the LGA.
  - **Vice-Chairman:** Assists the Chairman.
  - **Supervisory Councillors:** Heads of various departments (e.g., Education, Health, Works, Agriculture).
- **The Legislative Arm:**
  - **The Legislative Council:** Composed of **Councillors** elected from individual wards within the LGA.
  - **Functions:** They pass **by-laws** (local laws), approve the LGA's budget, and oversee the activities of the executive arm.
- **The Administrative Arm:**

- **Head of Local Government Administration (HLGA)/Secretary:** The most senior civil servant who oversees the day-to-day administration.
- **Departmental Heads:** Career civil servants who manage various departments.

### ***3.0 FUNCTIONS OF LOCAL GOVERNMENTS***

As outlined in the **Fourth Schedule of the 1999 Nigerian Constitution**, LGAs are responsible for:

1. **Infrastructure:** Construction and maintenance of roads, streets, drains, parks, and public conveniences.
2. **Health and Sanitation:** Provision and maintenance of health services; refuse disposal, and public sanitation.
3. **Education:** Establishment and maintenance of primary schools (in collaboration with the State Universal Basic Education Board - SUBEB).
4. **Revenue Collection:** Collection of rates, radio and television licenses, and other local taxes.
5. **Registration:** Registration of births, deaths, and marriages.
6. **Economic Development:** Control and regulation of markets, motor parks, bakeries, and restaurants.

### ***4.0 SOURCES OF REVENUE***

1. **External Sources:**
  - **Statutory Allocation:** Monthly funds from the Federation Account.
  - **Value Added Tax (VAT) Allocation:** A share of the national VAT pool.
  - **Grants:** From the Federal and State governments for specific projects.
2. **Internal Sources (Internally Generated Revenue - IGR):**
  - **Rates:** Tenement rates, market fees, and parking fees.
  - **Licenses:** Fees for marriage, bicycle, radio, and business licenses.
  - **Fines:** From courts for minor offenses.

## ***5.0 CHALLENGES FACING LOCAL GOVERNMENTS IN NIGERIA***

1. **Financial Inadequacy and Dependence:** Heavy reliance on federal allocations and very poor IGR generation.
2. **Political Interference:** State governments often usurp LGA functions and funds, undermining their autonomy.
3. **Corruption and Mismanagement:** Embezzlement of public funds and lack of accountability.
4. **Poor Personnel Management:** Lack of qualified staff, nepotism, and political appointments.
5. **Inadequate Infrastructure:** Many LGA secretariats lack basic facilities for effective service delivery.

## ***6.0 THE PATH TO REFORM: PROPOSED SOLUTIONS***

1. **Financial Autonomy:** Ensure statutory allocations are paid directly to LGAs and strengthen IGR collection mechanisms.
2. **Administrative Autonomy:** Reduce undue interference from state governments.
3. **Accountability and Transparency:** Implement regular audits, publish budgets and expenditure, and involve communities in monitoring projects.
4. **Capacity Building:** Train LGA staff and elected officials in modern public administration.
5. **Community Participation:** Encourage the formation of community development associations to partner with the LGA.

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## ***EVALUATION (WEEK 10)***

1. What is a Local Government Area (LGA), and why was this tier of government created in Nigeria?
2. Describe the structure of a typical LGA, naming the three arms and their key officers.
3. List five key functions of the Local Government as stated in the Nigerian constitution.

4. Explain the difference between statutory allocation and internally generated revenue (IGR). Why is a high IGR important for an LGA?
5. Identify three major challenges that hinder the effectiveness of LGAs in Nigeria and suggest one solution for each.

***ASSIGNMENT (WEEK 10)***

1. **Fieldwork/Research Project:** Investigate your own Local Government Area. a) Find out its name, the name of the current Chairman, and the number of wards it contains. b) Identify two recent projects (e.g., a constructed road, a renovated health center) executed by your LGA. c) Write a short report (300 words) on your findings, assessing the impact of one of these projects on the community.
  2. Write a formal letter to the Chairman of your LGA. In the letter, propose one specific, achievable project for your community and explain how it would benefit the residents.
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# SS1 GEOGRAPHY LESSON NOTES

## SECOND TERM

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### WEEK 1: MOUNTAINS

#### *1.0 DEFINITION AND SIGNIFICANCE OF MOUNTAINS*

A **mountain** is a natural elevation of the Earth's surface that rises abruptly to a summit, generally achieving an elevation of at least **600 meters (2,000 feet)** above the surrounding land. Mountains are not just impressive landforms; they are crucial regulators of global climate, water cycles, and centers of biodiversity.

#### *2.0 TYPES OF MOUNTAINS AND THEIR FORMATION PROCESSES*

Mountains are classified based on their mode of formation.

##### **2.1 Fold Mountains**

- **Formation Process:** Created by the **collision of tectonic plates**. The immense compressional forces cause layers of sedimentary rock to buckle and fold, much like a tablecloth being pushed together.
- **Characteristics:** Long, linear ranges with parallel ridges and valleys. They are the most common and often the tallest mountain types.
- **Examples:**
  - **Himalayas** (Asia: formed by the Indian plate colliding with the Eurasian plate)
  - **Andes** (South America)
  - **Alps** (Europe)
  - **Rockies** (North America)

##### **2.2 Block Mountains (Horst)**

- **Formation Process:** Formed by **tensional forces** in the Earth's crust. Parallel cracks or **faults** develop. The central block (horst) is uplifted or the surrounding blocks (grabens) sink down.

- **Characteristics:** Have steep, straight fronts and a flat top. They are often associated with **Rift Valleys** (the sunken blocks).
- **Examples:**
  - **Vosges Mountains** (France)
  - **Black Forest** (Germany)
  - **Sierra Nevada** (USA)
  - **Ruwenzori Mountains** (Uganda/DRC)

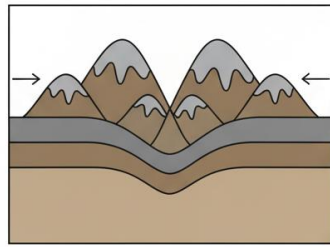
### 2.3 Volcanic Mountains

- **Formation Process:** Built up by the **extrusion of lava, ash, and other volcanic materials** from a vent in the Earth's crust.
- **Characteristics:** Typically conical in shape with a crater at the summit.
- **Examples:**
  - **Mount Kilimanjaro** (Tanzania)
  - **Mount Fuji** (Japan)
  - **Mount Cameroon** (Cameroon)
  - **Mauna Kea** (Hawaii)

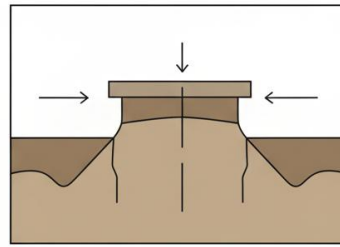
### 2.4 Residual Mountains (Mountains of Denudation)

- **Formation Process:** Formed not by internal forces, but by **differential weathering and erosion**. Softer rocks are worn away over millions of years, leaving behind more resistant rocks that stand as mountains.
- **Characteristics:** Often isolated and rugged.
- **Examples:**
  - **Aso Rock** (Nigeria)
  - **Zuma Rock** (Nigeria)
  - **The Olumo Rock** (Nigeria)
  - **Inselbergs** in Northern Nigeria

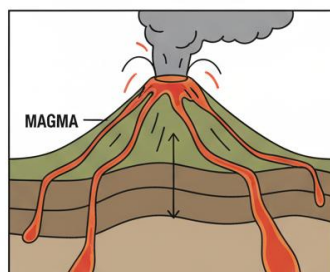
## MOUNTAIN FORMATION



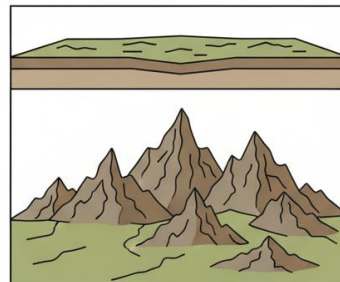
1. FOLD MOUNTAINS



2. BLOCK MOUNTAINS



3. VOLCANIC MOUNTAINS



4. RESIDUAL MOUNTAINS

**Illustration: Mountain Formation Diagrams** (Four simple diagrams showing: 1) Plates colliding to form fold mountains. 2) Tension causing faulting and uplift to form block mountains. 3) Magma erupting to form a volcanic mountain. 4) A plateau being eroded to form residual mountains.)

### 3.0 IMPORTANCE OF MOUNTAINS

3. **Water Towers:** They are the source of most of the world's major rivers (e.g., the Ganges from the Himalayas).
4. **Biodiversity Hotspots:** They host a wide variety of plants and animals, many of which are endemic.
5. **Tourism and Recreation:** Mountain climbing, hiking, skiing, and sightseeing generate significant revenue.
6. **Mineral Resources:** Many valuable minerals are mined in mountainous regions (e.g., tin on the Jos Plateau).

7. **Agricultural Terraces:** Steep slopes are terraced for farming (e.g., rice in the Philippines).
8. **Cultural and Spiritual Significance:** Many mountains are considered sacred sites.

#### ***4.0 MAJOR MOUNTAINS IN NIGERIA AND AFRICA***

- **In Nigeria:** The Jos Plateau, Obudu Plateau, Mambilla Plateau, and various inselbergs (Aso Rock, Zuma Rock) are classified as highlands, with the Mambilla Plateau being the highest.
  - **In Africa:** Mount Kilimanjaro (Tanzania), Mount Kenya (Kenya), Mount Stanley (Ruwenzori), Mount Cameroon.
- 

#### ***EVALUATION (WEEK 1)***

1. What is the minimum height generally used to define a mountain?
2. Explain, with the aid of a diagram, how fold mountains are formed.
3. Differentiate between a volcanic mountain and a residual mountain.
4. List four ways in which mountains are important to human activities.
5. Name two major highland areas in Nigeria and one famous mountain in Africa.

#### ***ASSIGNMENT (WEEK 1)***

1. Draw and label clear diagrams illustrating the formation of the four types of mountains.
  2. Research and write a one-page report on the Himalayas, focusing on their formation, highest peak, and their importance to the climate and people of Asia.
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## WEEK 2: LOWLANDS (PLAINS)

### 1.0 DEFINITION AND TYPES OF LOWLANDS

**Lowlands** (or Plains) are extensive areas of flat or gently rolling land, typically lying at low elevations (below 300 meters). They are formed by the deposition of sediments over long periods.

- **Types of Lowlands:**

- a. **Structural Plains:** Formed by the uplift of ancient sea floors (e.g., the Great Plains of the USA).
- b. **Depositional Plains:** Formed by the deposition of materials by rivers, wind, or glaciers.
  - **Alluvial Plains:** Formed by rivers (e.g., the Nile Delta in Egypt, the Niger-Benue Trough in Nigeria).
  - **Flood Plains:** The flat land on either side of a river that is subject to flooding.
  - **Coastal Plains:** Lowlands along the coast (e.g., the Coastal Plain of Lagos).
  - **Glacial Plains:** Formed by deposits from glaciers (e.g., the North European Plain).

### 2.0 FORMATION PROCESSES OF LOWLANDS

1. **Deposition by Rivers (Alluviation):** Rivers carry sediments (silt, sand, clay) from highlands and deposit them in their lower courses, forming fertile floodplains and deltas.
2. **Deposition by Wind (Aeolian Processes):** In arid regions, wind can deposit fine particles like loess to form plains.
3. **Deposition by Glaciers:** As glaciers melt, they drop their load of rock and debris, forming flat plains.
4. **Uplift of Sea Beds:** When parts of the ocean floor are raised above sea level, they form flat structural plains.

### ***3.0 IMPORTANCE OF LOWLANDS***

1. **Agriculture:** The deep, fertile soils are ideal for growing crops (e.g., rice in the Niger Delta, wheat in the Great Plains).
2. **Settlement and Transportation:** The flat terrain makes it easy to build cities, roads, and railways. Most of the world's major cities are on plains.
3. **Mineral Resources:** Many plains, especially deltas, are rich in resources like petroleum and natural gas (e.g., the Niger Delta).
4. **Industrial Development:** Flat land and easy transportation facilitate the establishment of industries.

### ***4.0 PROBLEMS ASSOCIATED WITH LOWLANDS***

1. **Flooding:** Being flat and often near rivers, they are highly susceptible to seasonal and catastrophic floods.
2. **Water-logging:** Poor drainage can make the soil saturated, limiting its use for agriculture or construction.
3. **Disease:** Stagnant water in lowland areas provides breeding grounds for mosquitoes, leading to malaria and other water-borne diseases.

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### ***EVALUATION (WEEK 2)***

1. Define the term "lowland" and name three types of lowlands.
2. Describe how river deposition leads to the formation of a floodplain.
3. Explain three reasons why lowlands are more densely populated than highland areas.
4. Identify two problems faced by people living in lowland areas.

### ***ASSIGNMENT (WEEK 2)***

1. Draw a simple diagram showing a cross-section of a river from its source to its mouth, labeling the upper course (highlands), middle course, and lower course (lowlands/floodplain).

2. Write short notes on the **Niger Delta**, describing its location, how it was formed, its economic importance, and one major environmental challenge it faces.
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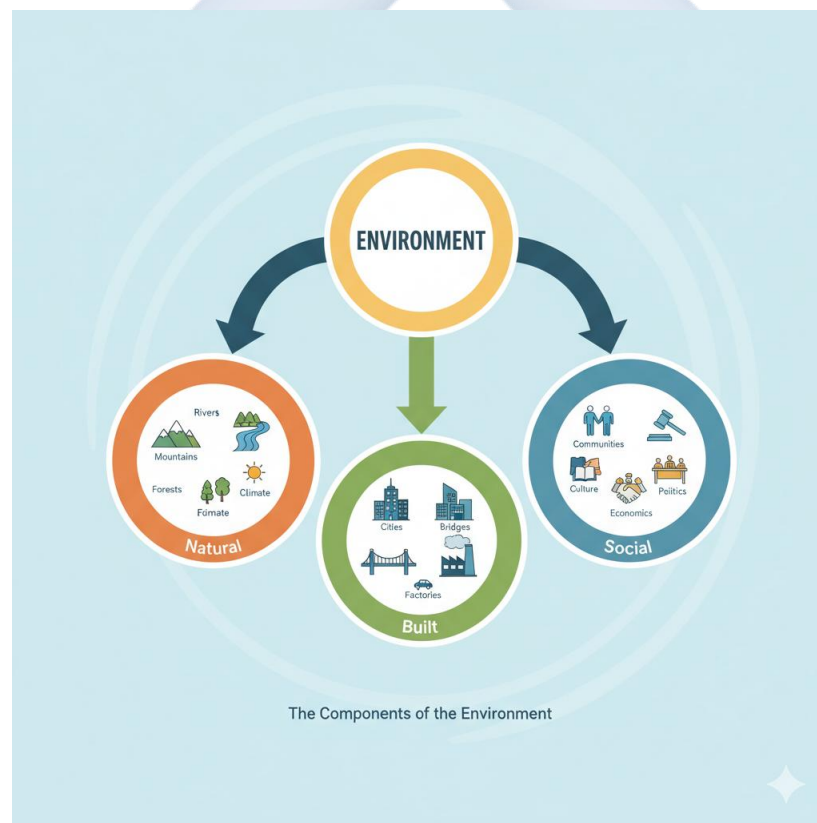


## WEEK 3: THE ENVIRONMENT

### 1.0 DEFINITION AND COMPONENTS OF THE ENVIRONMENT

The **environment** is the sum total of all living and non-living things, conditions, and influences that surround and affect an organism. It is a complex, interconnected system.

- **Components of the Environment:**
  - a. **Natural (Physical) Environment:** All naturally occurring things (land, water, air, plants, animals).
  - b. **Built (Human-Made) Environment:** Everything created by humans (buildings, roads, farms, factories).
  - c. **Social/Cultural Environment:** Human institutions, beliefs, customs, and economic systems.



**Illustration: The Components of the Environment** (A diagram with "ENVIRONMENT" at the center, connected to three circles: "Natural," "Built," and "Social," with examples in each circle.)

## ***2.0 THE ECOSYSTEM CONCEPT***

An **ecosystem** is a community of living organisms (biotic components) in conjunction with the non-living components (abiotic components) of their environment, interacting as a system. (e.g., a pond, a forest, a savanna).

## ***3.0 HUMAN IMPACT ON THE ENVIRONMENT***

- **Negative Impacts (Environmental Degradation):**
  - a. **Pollution:** The introduction of harmful materials into the environment (air pollution from industries, water pollution from sewage and oil spills, land pollution from plastic waste).
  - b. **Deforestation:** The large-scale clearing of forests for agriculture, logging, or urbanization, leading to soil erosion, loss of biodiversity, and climate change.
  - c. **Desertification:** The process by which fertile land becomes desert, typically as a result of drought, deforestation, or inappropriate agriculture (a major problem in Northern Nigeria).
  - d. **Global Warming:** The long-term heating of Earth's climate system due to human activities, primarily the burning of fossil fuels, which increases heat-trapping greenhouse gas levels in the atmosphere.
- **Positive Impacts (Environmental Conservation):**
  - a. **Afforestation and Reforestation:** Planting trees to create new forests or restore old ones.
  - b. **Recycling:** Converting waste into reusable material.
  - c. **Establishment of Protected Areas:** Creating national parks and game reserves to protect wildlife and ecosystems.
  - d. **Use of Renewable Energy:** Shifting from fossil fuels to solar, wind, and hydroelectric power.

#### ***4.0 SUSTAINABLE DEVELOPMENT***

**Sustainable Development** is development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It balances economic growth with environmental protection and social well-being.

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#### ***EVALUATION (WEEK 3)***

1. Define the term "environment" and list its three main components.
2. What is an ecosystem? Give one example.
3. Explain three negative impacts of human activities on the environment.
4. Suggest three ways in which we can protect and conserve our environment for future generations.
5. What is the meaning of "sustainable development"?

#### ***ASSIGNMENT (WEEK 3)***

1. Create a poster aimed at creating awareness about one environmental issue in your community (e.g., improper waste disposal, deforestation, pollution). The poster should illustrate the problem and suggest solutions.
  2. Write a short essay on: "The Role of Students in Environmental Protection."
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## WEEK 4: WEATHER

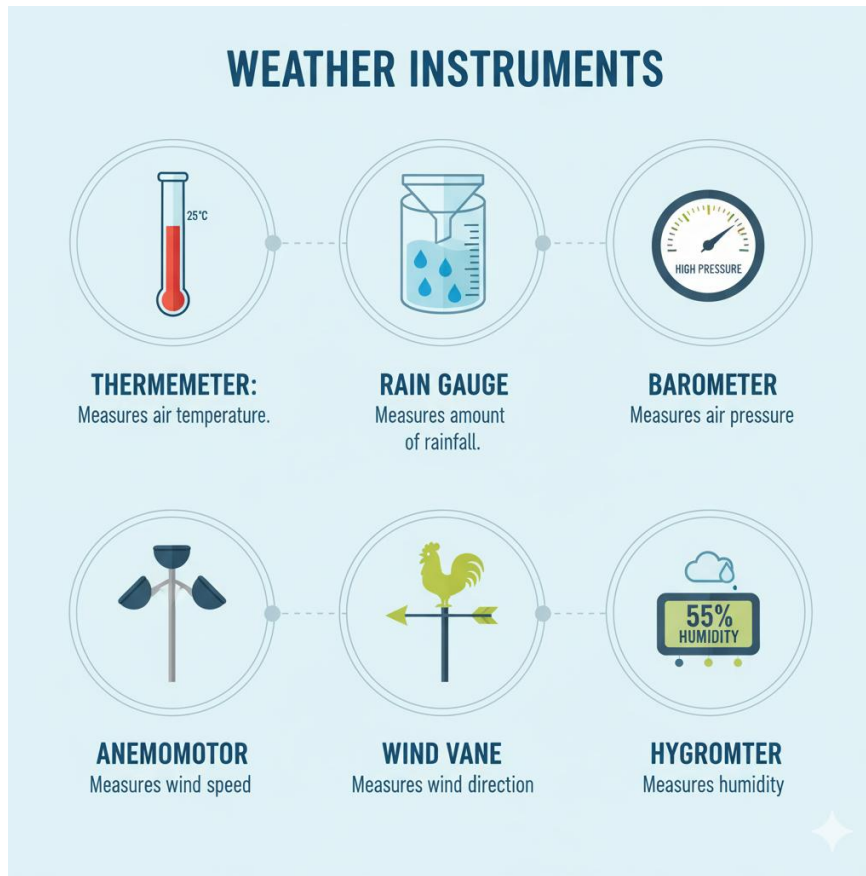
### 1.0 DEFINITION: WEATHER vs. CLIMATE

- **Weather:** The **day-to-day** state of the atmosphere at a specific place and time, with respect to heat, dryness, sunshine, wind, rain, etc. It is **short-term** and **changes frequently**.
  - *Example:* "It is sunny and 32°C in Lagos today."
- **Climate:** The **average weather conditions** of a place, calculated over a long period (usually **30-35 years**). It is **long-term** and **general**.
  - *Example:* "Nigeria has a tropical climate with distinct wet and dry seasons."

### 2.0 ELEMENTS OF WEATHER

These are the measurable components that define the weather.

1. **Temperature:** The degree of hotness or coldness of the air (measured in °C or °F with a **Thermometer**).
2. **Precipitation:** Any form of water that falls from the atmosphere (rain, snow, hail). Measured with a **Rain Gauge**.
3. **Humidity:** The amount of water vapor in the air. Measured with a **Hygrometer**.
4. **Atmospheric Pressure:** The force exerted by the weight of the air above. Measured with a **Barometer**.
5. **Wind:** The movement of air from high pressure to low pressure areas.
  - **Wind Speed:** Measured with an **Anemometer**.
  - **Wind Direction:** Measured with a **Wind Vane**.
6. **Cloud Cover:** The fraction of the sky obscured by clouds. Observed in **Oktas** (eighths).
7. **Sunshine:** The duration of sunlight. Measured with a **Sunshine Recorder**.



**Illustration: Weather Instruments** (A labeled diagram showing the six main weather instruments: Thermometer, Rain Gauge, Barometer, Anemometer, Wind Vane, and Hygrometer.)

### 3.0 WEATHER STATIONS AND FORECASTING

A **weather station** is a facility with instruments and equipment to observe and record atmospheric conditions. Data from thousands of stations, along with satellite and radar information, is used by **meteorologists** to make **weather forecasts**.

### 4.0 IMPORTANCE OF STUDYING WEATHER

1. **Agriculture:** Farmers rely on weather forecasts to decide when to plant, irrigate, and harvest.
2. **Aviation and Shipping:** Pilots and sailors need weather information for safe travel.
3. **Disaster Preparedness:** Warnings for storms, floods, and hurricanes save lives and property.

4. **Daily Activities:** Influences what we wear, our travel plans, and outdoor events.
  5. **Health:** Helps in predicting the spread of certain diseases and managing heatwaves.
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#### ***EVALUATION (WEEK 4)***

1. Differentiate between "weather" and "climate."
2. List four elements of weather and the instrument used to measure each one.
3. Why is a barometer an important weather instrument?
4. Explain three reasons why accurate weather forecasting is important for a country like Nigeria.

#### ***ASSIGNMENT (WEEK 4)***

1. Draw and label the six main weather instruments.
  2. **Practical Project:** Create a simple weather diary for one week. Each day, record your observations for:
    - Date
    - General Weather (Sunny, Cloudy, Rainy)
    - Estimated Temperature (Hot, Warm, Cool)
    - Wind Conditions (Calm, Breezy, Windy)
    - Any Rainfall (None, Light, Heavy)
-

## WEEK 6: CLIMATE I

### 1.0 DEFINITION AND DISTINCTION FROM WEATHER

- **Climate** is the **long-term** pattern of weather in a particular area, typically averaged over a period of **30 years or more**. It represents the atmospheric conditions that are typical for a region.
- While weather changes hourly or daily, climate describes the **expected conditions** for a location based on historical data.

#### Key Differences Table:

| Feature               | Weather                  | Climate                                                 |
|-----------------------|--------------------------|---------------------------------------------------------|
| <b>Time Period</b>    | Short-term (hours, days) | Long-term (years, decades)                              |
| <b>Nature</b>         | Specific, changing       | General, stable pattern                                 |
| <b>Description</b>    | "It is raining today."   | "This region has a rainy season from April to October." |
| <b>Predictability</b> | Forecast for days ahead  | Projected trends over years                             |

### 2.0 ELEMENTS AND CONTROLS OF CLIMATE

The **elements** of climate are the same as those of weather (temperature, precipitation, humidity, etc.), but they are studied as long-term averages.

The **controls** of climate are the factors that cause different places to have different climates:

1. **Latitude:** The most important control. It determines the amount of solar energy received. Areas near the equator (low latitudes) are hot, while polar regions (high latitudes) are cold.
2. **Altitude:** Temperature generally decreases with height (approximately 6.5°C for every 1000m). Highlands are cooler than lowlands at the same latitude.



3. **Distance from the Sea (Continentality):** Land heats up and cools down faster than water. Coastal areas have milder, more moderate climates (maritime climate), while interior regions experience extreme temperatures (continental climate).
4. **Ocean Currents:** Warm currents raise the temperature of coastal areas (e.g., the Gulf Stream warms NW Europe), while cold currents lower it (e.g., the Benguela Current cools the coast of Namibia).
5. **Prevailing Winds:** Winds carry the temperature and moisture characteristics of their source region.
6. **Aspect:** In mountainous regions, south-facing slopes (in the Northern Hemisphere) receive more direct sunlight and are warmer than north-facing slopes.

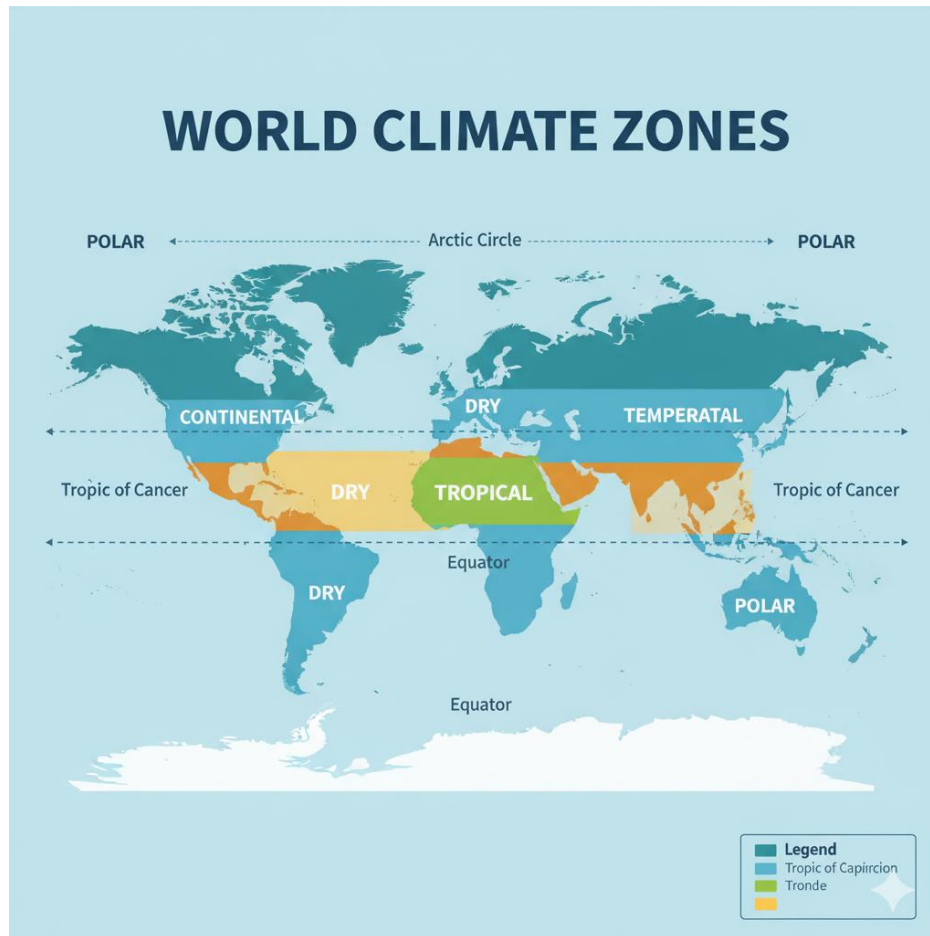
### ***3.0 CLASSIFICATION OF CLIMATE: THE KÖPPEN SYSTEM***

The Köppen climate classification system is the most widely used. It categorizes climates based on temperature and precipitation patterns. Major groups include:

- **Tropical (A):** Found near the equator. Consistently high temperatures and high rainfall year-round (e.g., Southern Nigeria).
- **Dry (B):** Characterized by low precipitation. Includes deserts (BWh/BWk) and semi-arid steppes (BSh/BSk) (e.g., Northern Nigeria).
- **Temperate (C):** Warm summers and mild winters with moderate rainfall (e.g., Mediterranean Europe).
- **Continental (D):** Found in the interior of continents, with warm summers and very cold winters.
- **Polar (E):** Permanently cold, with temperatures never exceeding 10°C.



#### 4.0 THE EARTH'S MAJOR CLIMATE ZONES



**Illustration: World Climate Zones Map** (A simplified world map showing the general belts of Tropical, Dry, Temperate, Continental, and Polar climates, with the Equator, Tropics, and Polar Circles marked.)

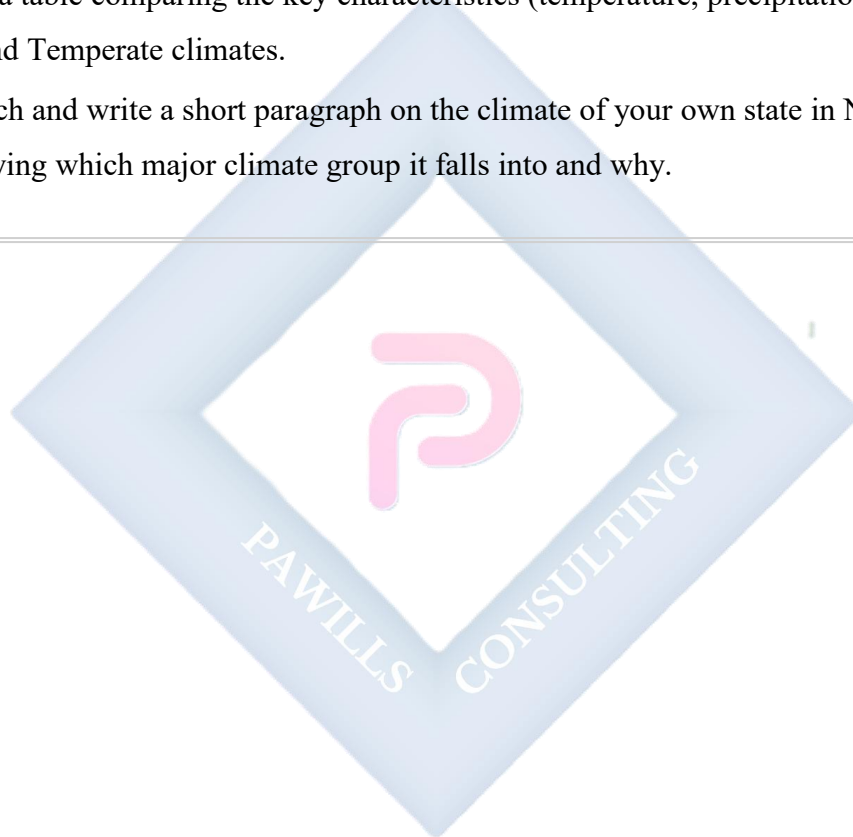
1. **The Tropics (Low Latitudes):** Between the Tropic of Cancer and Capricorn. High temperatures year-round.
2. **The Temperate Zones (Mid-Latitudes):** Between the tropics and the polar circles. Distinct seasonal changes.
3. **The Polar Zones (High Latitudes):** Within the Arctic and Antarctic Circles. Extremely cold, with midnight sun and polar nights.

### ***EVALUATION (WEEK 6)***

1. What is the fundamental difference between weather and climate?
2. Explain how latitude and altitude are two of the most important controls of climate.
3. If a city is located at a high altitude near the equator, what would you expect its climate to be like compared to a coastal city at the same latitude?
4. Name the five major groups in the Köppen climate classification system.

### ***ASSIGNMENT (WEEK 6)***

1. Create a table comparing the key characteristics (temperature, precipitation) of Tropical, Dry, and Temperate climates.
  2. Research and write a short paragraph on the climate of your own state in Nigeria, identifying which major climate group it falls into and why.
- 



## WEEK 8: NIGERIA: LOCATION AND POSITION

### 1.0 ABSOLUTE AND RELATIVE LOCATION

- **Absolute Location:** The exact position on Earth's surface using a grid system (latitude and longitude).
  - **Nigeria's Coordinates:**
    - **Latitude:** Between **4°N and 14°N** of the Equator.
    - **Longitude:** Between **3°E and 15°E** of the Prime Meridian.
  - **Implication:** This places Nigeria entirely within the **Northern Hemisphere** and the **Eastern Hemisphere**.
- **Relative Location:** The location of a place in relation to other places.
  - Nigeria is located in **West Africa**, on the **Gulf of Guinea** (part of the Atlantic Ocean).

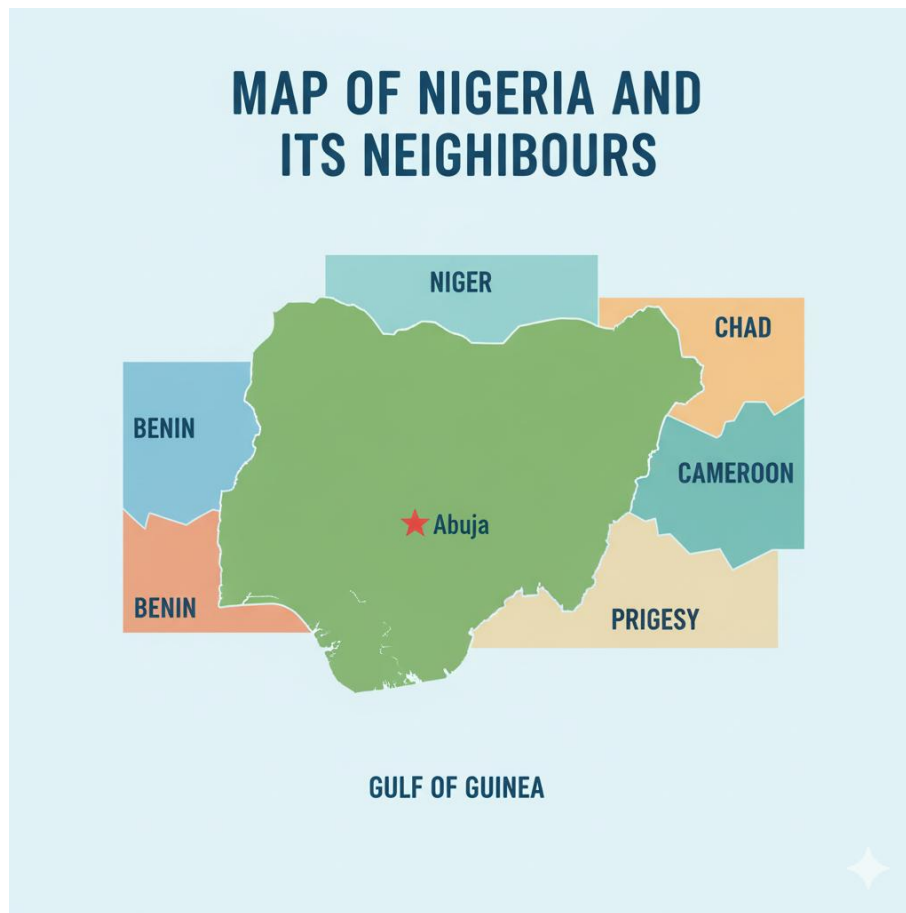
### 2.0 SIZE AND EXTENT

- Nigeria has a total area of approximately **923,768 square kilometers**, making it one of the largest countries in Africa.
- The latitudinal extent (4°N to 14°N) is about **1,100 km**.
- The longitudinal extent (3°E to 15°E) is about **1,200 km**.

### 3.0 NEIGHBOURING COUNTRIES AND BOUNDARIES

Nigeria shares land borders with four countries and a maritime border:

- **North: Niger Republic** (longest border).
- **East: Cameroon**.
- **West: Republic of Benin**.
- **Northeast: Chad** (across Lake Chad).
- **South: Gulf of Guinea** (Atlantic Ocean).



**Illustration: Map of Nigeria and its Neighbours** *(A clear outline map of Nigeria showing the four bordering countries and the Gulf of Guinea, with the capital city, Abuja, marked.)*

#### **4.0 GEOPOLITICAL AND ECONOMIC SIGNIFICANCE OF NIGERIA'S POSITION**

1. **Regional Leadership:** Its large size, population, and economy make Nigeria a natural leader in West Africa and within organizations like **ECOWAS** (Economic Community of West African States).
2. **Maritime Access:** The coastline on the Gulf of Guinea provides access to international sea routes, facilitating trade through major ports like Lagos (Apapa, Tin Can), Port Harcourt, and Calabar.
3. **Cross-Border Trade:** Its position encourages trade and cultural exchange with neighboring countries.

4. **Climate Variation:** The north-south extent results in a climatic gradient from the humid south to the arid north, supporting a diversity of agricultural products.
  5. **Security Challenges:** Porous borders can pose challenges, such as the movement of insurgent groups and smuggling.
- 

### ***EVALUATION (WEEK 8)***

1. State the latitudinal and longitudinal extent of Nigeria.
2. Name the four countries that share a land border with Nigeria.
3. Explain the difference between absolute location and relative location, using Nigeria as an example.
4. What are two advantages of Nigeria's location on the Gulf of Guinea?

### ***ASSIGNMENT (WEEK 8)***

1. Draw a neat outline map of Nigeria. On it, mark and label the following:
    - The four neighbouring countries.
    - The Gulf of Guinea.
    - The capital city, Abuja.
    - The approximate lines of latitude 4°N, 8°N, and 12°N.
  2. Write a short essay (250 words) on the topic: "The Implications of Nigeria's Size and Location for its Economic Development."
-

## WEEK 9: PHYSICAL SETTING OF NIGERIA

### 1.0 MAJOR PHYSICAL REGIONS (RELIEF)

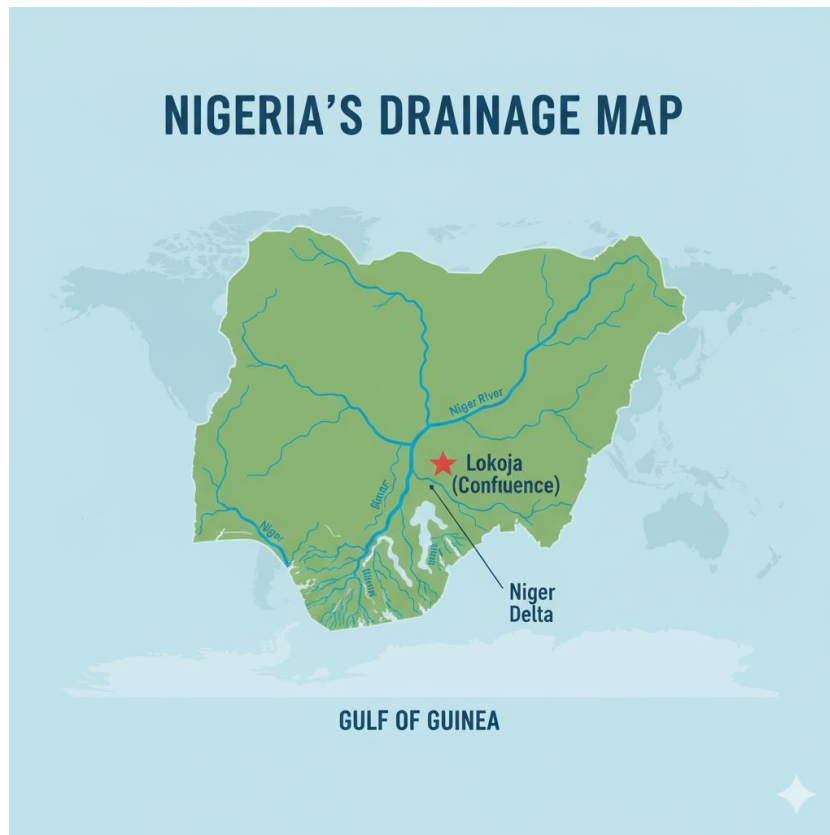
Nigeria's landscape can be divided into two broad relief regions:

1. **The Highlands:** Areas over 300m above sea level.
  - **The Eastern Highlands:** Includes the Jos Plateau (average 1,200m), Adamawa Highlands, and the Obudu Plateau. Rich in minerals like tin.
  - **The Western Highlands:** Includes the Yoruba Highlands and the Idanre Hills.
  - **The North-Central Highlands:** Includes the Biu Plateau.
2. **The Lowlands:** Areas below 300m.
  - **The Sokoto Plains:** In the northwest.
  - **The Niger-Benue Trough:** The broad valley through which the Niger and Benue rivers flow.
  - **The Chad Basin:** A low-lying area in the northeast.
  - **The Coastal Plains:** A broad belt along the entire coast, including the Niger Delta.

### 2.0 DRAINAGE: RIVERS AND THEIR PATTERNS

Nigeria is drained mainly by the **River Niger** and its major tributary, the **River Benue**.

- **River Niger:** Enters Nigeria in the northwest and flows south to its delta in the Gulf of Guinea. It is Nigeria's longest river.
- **River Benue:** Rises in Cameroon and flows west to meet the Niger at **Lokoja** (the confluence town).
- **Other Major Rivers:** Kaduna, Gongola, Cross River, Sokoto, Ogun, and Imo.
- **Major Dams:** Kainji Dam (on the Niger), Shiroro Dam (on the Kaduna), and Kiri Dam (on the Gongola). They are used for **Hydroelectric Power (HEP)** generation, irrigation, and fishing.



**Illustration: Nigeria's Drainage Map** *(A map showing the major rivers of Nigeria, highlighting the Niger and Benue rivers and their confluence at Lokoja, and the Niger Delta.)*

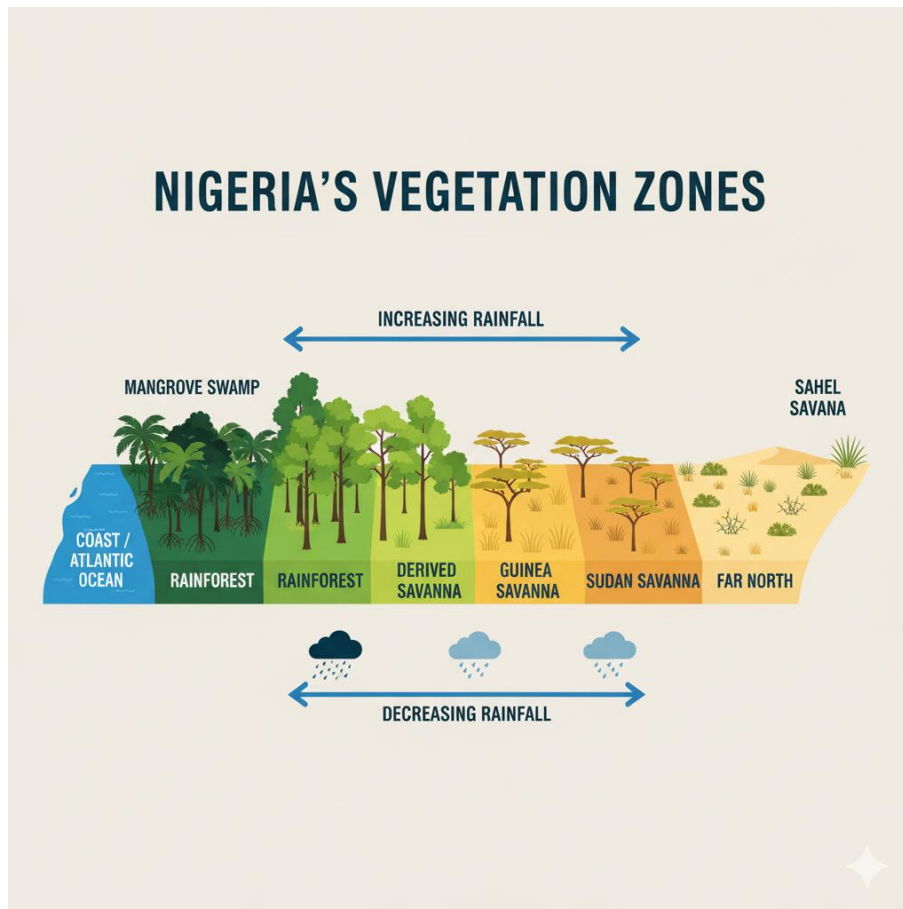
### 3.0 VEGETATION ZONES

Nigeria's vegetation changes from south to north, reflecting the gradient in rainfall.

1. **Mangrove Swamp Forest:** Found in the coastal belt of the Niger Delta. Salt-tolerant trees; important for fisheries and as a nursery for marine life.
2. **Tropical Rainforest:** South of the Niger-Benue trough. Dense, multi-layered forest with evergreen trees. Rich in biodiversity (timber, cocoa, rubber).
3. **Guinea Savannah (Parkland Savannah):** Middle belt. A mix of tall grasses and scattered trees (e.g., shea butter, locust bean). Suitable for crops like yam, maize, and groundnut.
4. **Sudan Savannah:** Northern Nigeria. Shorter grasses and fewer, shorter trees. The main grain-producing region (millet, sorghum, cowpea).



5. **Sahel Savannah:** Extreme northeast. Semi-desert conditions with short, scanty grasses and thorny shrubs. Vulnerable to desertification.
6. **Montane Vegetation:** Found on the highlands (Jos, Obudu). Grasses and scattered trees adapted to cooler temperatures.



**Illustration: Nigeria's Vegetation Zones** (A cross-sectional diagram from the coast to the far north, showing the succession of vegetation zones and how they correlate with decreasing rainfall.)

#### **4.0 INTERRELATIONSHIP: PHYSICAL ENVIRONMENT AND HUMAN ACTIVITIES**

- **Rainforest Zone:** Timber logging, cash crop cultivation (cocoa, oil palm).
- **Savannah Zones:** Cereal crop farming, cattle rearing.
- **Niger Delta:** Fishing, oil and gas exploration.
- **Jos Plateau:** Tin mining, temperate vegetable farming.
- **Coastal Areas:** Port activities, salt extraction, tourism.



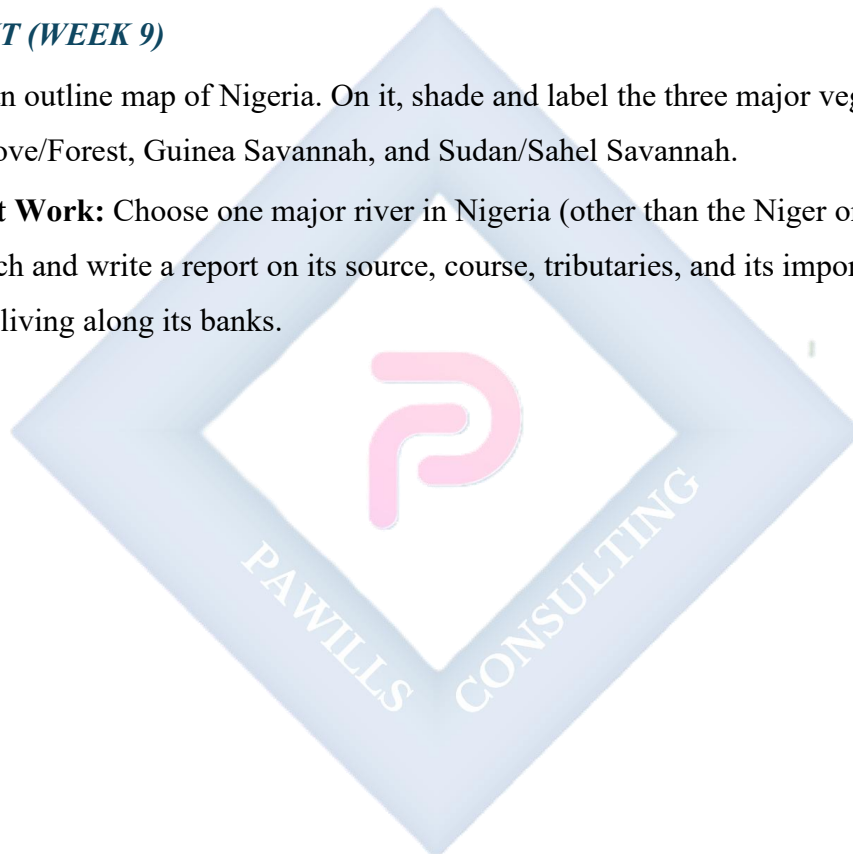
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### ***EVALUATION (WEEK 9)***

1. Name the two major highland areas in Nigeria and one characteristic of each.
2. What is the significance of Lokoja in Nigeria's drainage system?
3. List the four main vegetation zones of Nigeria from south to north.
4. Explain how the natural vegetation of a region influences the type of agricultural activity practiced there.

### ***ASSIGNMENT (WEEK 9)***

1. Draw an outline map of Nigeria. On it, shade and label the three major vegetation zones: Mangrove/Forest, Guinea Savannah, and Sudan/Sahel Savannah.
2. **Project Work:** Choose one major river in Nigeria (other than the Niger or Benue). Research and write a report on its source, course, tributaries, and its importance to the people living along its banks.



# SS1 GEOGRAPHY LESSON NOTES

## THIRD TERM

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### WEEK 1: POPULATION OF NIGERIA

#### 1.0 POPULATION CONCEPTS AND DEFINITIONS

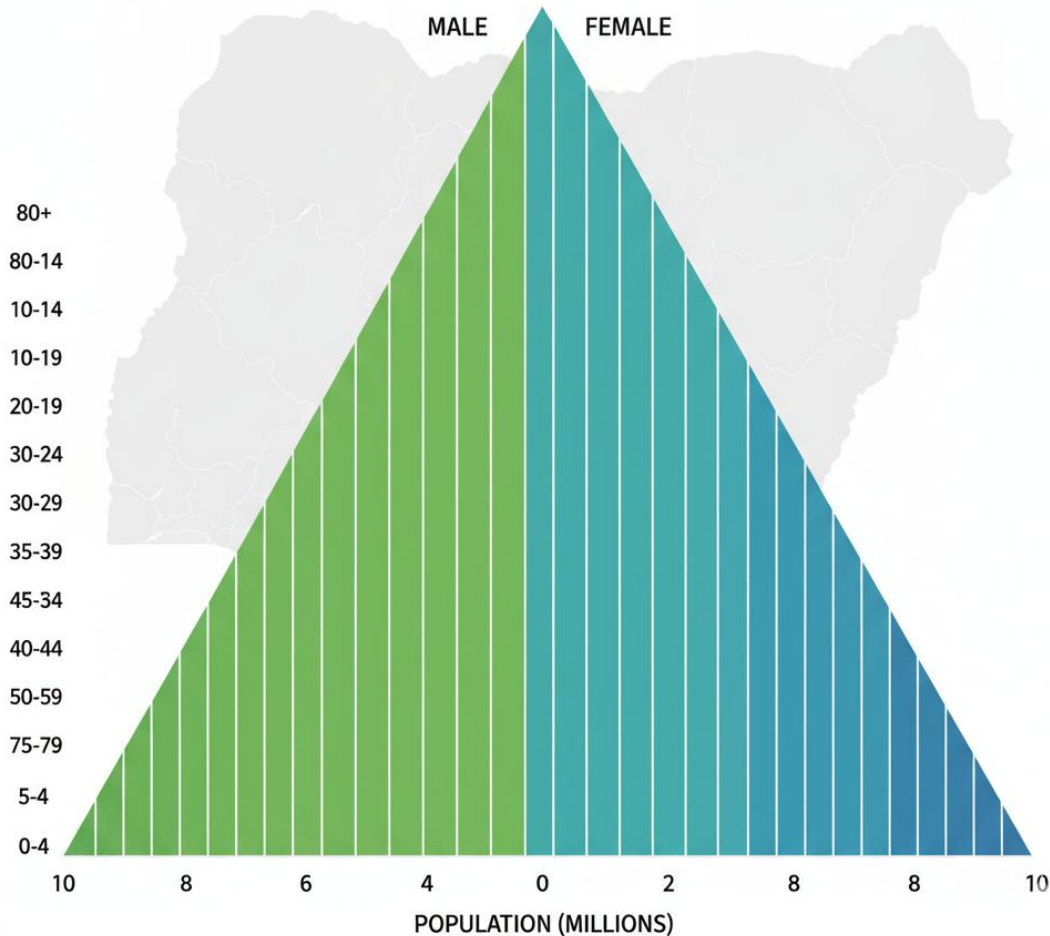
- **Population:** The total number of people living in a defined geographic area at a specific time.
- **Population Distribution:** The pattern of where people live within an area (e.g., clustered, dispersed).
- **Population Density:** The number of people per unit area (usually per square kilometer). Calculated as: **Total Population ÷ Total Land Area.**
- **Population Census:** An official count of the population, typically recording details like age, sex, and occupation. Nigeria's last official census was in 2006.

#### 2.0 POPULATION SIZE, GROWTH, AND STRUCTURE

- **Current Size:** Nigeria is the **most populous country in Africa** and the **sixth most populous in the world**, with an estimated population of over **220 million people** (2023 estimates).
- **Growth Rate:** Nigeria has a high population growth rate of approximately **2.6% per annum**. This is due to a high birth rate and a declining death rate.
- **Population Structure (Age-Sex Pyramid):** Nigeria has a **youthful population**. Over 60% of the population is under the age of 25. This creates a broad-based population pyramid.
  - **Implications:** High dependency ratio, pressure on educational facilities, and a large future workforce.

# NIGERIA'S POPULATION PYRAMID

NIGERIA: 2023



**Illustration: Nigeria's Population Pyramid** *(A diagram showing a typical expansive pyramid with a very wide base, tapering to a narrow top, indicating a high proportion of young people.)*

## 3.0 FACTORS AFFECTING POPULATION DISTRIBUTION IN NIGERIA

Population is unevenly distributed across Nigeria. Dense populations are found in the southwest, southeast, and around the Niger-Benue valley, while the northeast and some central areas are sparsely populated.

### 9. Physical Factors:

- **Relief:** Lowlands and plains (e.g., coastal areas, Niger-Benue valley) are more densely populated than highlands (e.g., Jos Plateau) which are cooler but less accessible.
- **Climate:** Areas with moderate rainfall and temperature (e.g., the Guinea Savanna) are more favorable than very wet (coastal) or very dry (Sahel) regions.
- **Soil Fertility:** Areas with fertile soil for agriculture (e.g., the Jos Plateau, river valleys) attract settlement.
- **Water Supply:** Proximity to rivers and lakes (e.g., the Niger, Benue) encourages settlement.

#### 10. Human and Economic Factors:

- **Industrialization and Employment:** Urban centers like Lagos, Port Harcourt, and Kano have high population densities due to job opportunities.
- **Transportation Networks:** Areas with good road, rail, and river transport are more accessible and attract people.
- **Social Amenities:** Availability of schools, hospitals, and electricity makes an area attractive.
- **Historical Factors:** Some ancient cities (e.g., Kano, Benin) have been population centers for centuries.
- **Government Policy:** The creation of a new state capital (e.g., Abuja) can attract population.

### ***4.0 IMPLICATIONS OF NIGERIA'S POPULATION CHARACTERISTICS***

- **Positive Implications:**

- a. **Large Labour Force:** Provides manpower for economic development.
- b. **Large Market:** A large population creates a big domestic market for goods and services.
- c. **Diversity:** A large population often means cultural and skill diversity.

- **Negative Implications (Challenges):**

- a. **Pressure on Resources:** Overpopulation can lead to overuse of land, water, and forest resources.
  - b. **Unemployment and Underemployment:** The economy may not create jobs fast enough for the growing labour force.
  - c. **Pressure on Social Amenities:** Overcrowding in schools, hospitals, and inadequate housing (leading to slums).
  - d. **Environmental Degradation:** Increased pollution, deforestation, and waste management problems.
  - e. **Food Shortage:** Agricultural production may not keep pace with population growth.
- 

### ***EVALUATION (WEEK 1)***

1. Define the terms "population density" and "population distribution."
2. Why is Nigeria's population described as "youthful," and what are two implications of this?
3. Explain three physical factors that influence where people choose to live in Nigeria.
4. List two positive and two negative consequences of Nigeria's rapidly growing population.

### ***ASSIGNMENT (WEEK 1)***

1. Using a map of Nigeria, identify and list three states with high population density and three states with low population density. Suggest reasons for the differences you observe.
  2. Calculate the population density of your Local Government Area (LGA) if the population is 250,000 and the land area is 500 square kilometers. Show your working.
-

## WEEK 2: RESOURCES OF NIGERIA

### 1.0 DEFINITION AND CLASSIFICATION OF RESOURCES

A **resource** is any physical material, substance, or human attribute that can be used to satisfy human needs and wants.

- **Classification:**

- a. **Based on Origin:**

- **Natural Resources:** Provided by nature (e.g., crude oil, water, forests).
    - **Human Resources:** The skills, knowledge, and labour of people.
    - **Capital Resources:** Human-made tools and equipment used in production (e.g., machinery, roads).

- b. **Based on Renewability:**

- **Renewable Resources:** Can be replenished naturally over time (e.g., solar energy, wind, forests, water).
    - **Non-Renewable Resources:** Exist in finite quantities and cannot be replaced once exhausted (e.g., crude oil, natural gas, coal, tin).

### 2.0 MAJOR RESOURCES IN NIGERIA AND THEIR DISTRIBUTION

Nigeria is endowed with a wide variety of natural resources.

#### Table of Major Resources and Their Locations:

| Resource                           | Type          | Major Locations in Nigeria                      |
|------------------------------------|---------------|-------------------------------------------------|
| <b>Crude Oil &amp; Natural Gas</b> | Non-Renewable | Niger Delta (Rivers, Bayelsa, Delta, Akwa Ibom) |
| <b>Coal</b>                        | Non-Renewable | Enugu, Benue, Kogi                              |
| <b>Tin &amp; Columbite</b>         | Non-Renewable | Jos Plateau                                     |
| <b>Iron Ore</b>                    | Non-Renewable | Itakpe (Kogi), Agbaja (Kogi)                    |
| <b>Limestone</b>                   | Non-Renewable | Ewekoro (Ogun), Ukpilla (Edo), Sokoto           |

| Resource         | Type                        | Major Locations in Nigeria    |
|------------------|-----------------------------|-------------------------------|
| <b>Gold</b>      | Non-Renewable               | Ilesa (Osun), Zamfara, Kaduna |
| <b>Cocoa</b>     | Renewable<br>(Agricultural) | Ondo, Osun, Oyo, Ekiti        |
| <b>Groundnut</b> | Renewable<br>(Agricultural) | Kano, Kaduna, Katsina         |
| <b>Palm Oil</b>  | Renewable<br>(Agricultural) | Imo, Abia, Akwa Ibom, Rivers  |
| <b>Rubber</b>    | Renewable<br>(Agricultural) | Edo, Delta, Cross River       |
| <b>Timber</b>    | Renewable (Forest)          | Cross River, Ondo, Ogun       |

**Illustration: Nigeria's Mineral Resources Map** *(A map of Nigeria showing the locations of the key mineral and agricultural resources.)*





### ***3.0 IMPORTANCE OF RESOURCES TO THE NIGERIAN ECONOMY***

1. **Source of Revenue:** Resources like crude oil provide the majority of Nigeria's foreign exchange earnings and government revenue.
2. **Industrial Raw Materials:** Limestone for cement, iron ore for steel, rubber for tires, etc.
3. **Employment Generation:** Mining, farming, and processing industries create jobs.
4. **Energy Production:** Natural gas, coal, and hydropower (from water resources) are used for electricity generation.
5. **Food Supply:** Agricultural resources provide food for the population.

### ***4.0 PROBLEMS OF RESOURCE DEVELOPMENT AND MANAGEMENT***

1. **Over-dependence on Crude Oil:** The economy is vulnerable to fluctuations in global oil prices ("Dutch Disease").
2. **Environmental Pollution:** Oil spills in the Niger Delta and gas flaring have devastated the environment.
3. **Underutilization of Other Resources:** Many solid minerals and agricultural potentials are not fully developed.
4. **Corruption and Mismanagement:** Revenue from resources is often not used effectively for national development.
5. **Conflict:** Competition over resource control can lead to conflicts, as seen in the Niger Delta.

### ***5.0 THE NEED FOR ECONOMIC DIVERSIFICATION***

There is an urgent need for Nigeria to diversify its economy away from oil by:

- Revitalizing agriculture for export and food security.
- Developing the solid minerals sector.
- Investing in manufacturing and tourism.
- Promoting small and medium-sized enterprises (SMEs).

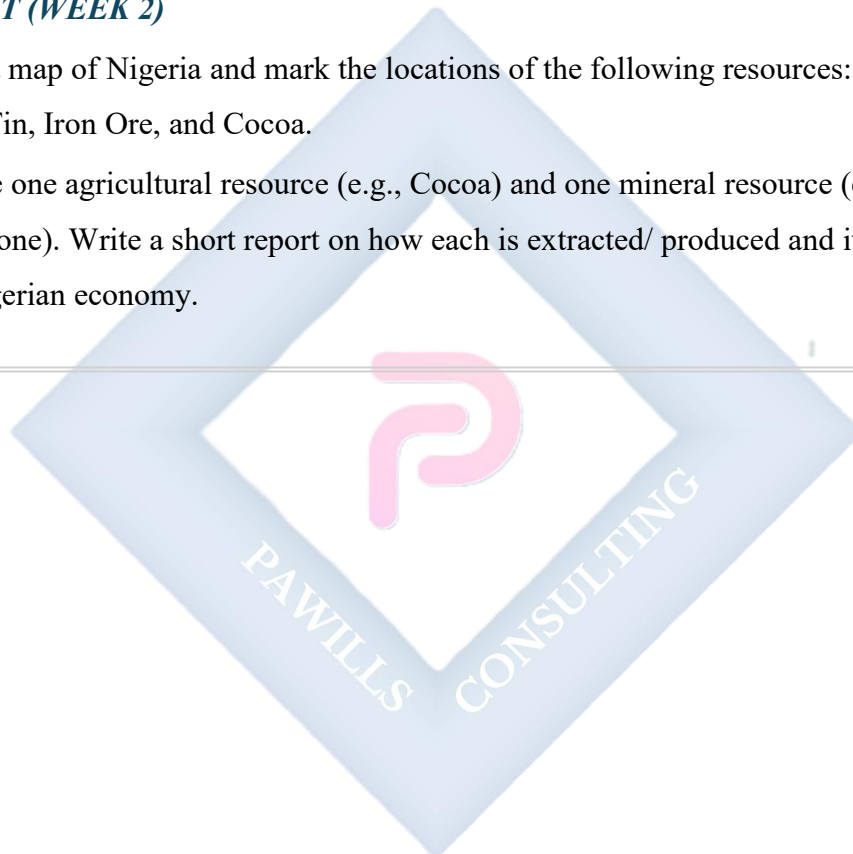


### ***EVALUATION (WEEK 2)***

1. Differentiate between renewable and non-renewable resources, giving two examples of each from Nigeria.
2. Name the mineral resource found in the Niger Delta and two solid minerals found in the Jos Plateau.
3. Explain three problems associated with the development of resources in Nigeria.
4. Why is economic diversification important for Nigeria?

### ***ASSIGNMENT (WEEK 2)***

1. Draw a map of Nigeria and mark the locations of the following resources: Crude Oil, Coal, Tin, Iron Ore, and Cocoa.
2. Choose one agricultural resource (e.g., Cocoa) and one mineral resource (e.g., Limestone). Write a short report on how each is extracted/ produced and its importance to the Nigerian economy.



## WEEK 3: BASIC CONCEPTS IN MAP READING

### 1.0 DEFINITION AND COMPONENTS OF A MAP

A **map** is a symbolic, scaled-down representation of the Earth's surface or part of it on a flat surface.

- **Essential Components of a Map:**
  - a. **Title:** Tells you what the map represents.
  - b. **Scale:** Shows the relationship between map distance and real ground distance.
  - c. **Key (Legend):** Explains the meaning of the symbols and colors used on the map.
  - d. **North Arrow (Compass Rose):** Shows the direction of North on the map.
  - e. **Grid System:** A network of lines (latitude and longitude) used to locate places.
  - f. **Border (Neatline):** The frame that encloses the map.

### 2.0 MAP SCALE

The scale is crucial for understanding the size and distance on a map. It can be represented in three ways:

1. **Statement Scale:** Expressed in words (e.g., "one centimeter represents one kilometer").
2. **Representative Fraction (R.F.):** Expressed as a ratio (e.g., 1:100,000). This means 1 unit on the map equals 100,000 of the same units on the ground.
3. **Linear (Graphical) Scale:** A line divided into segments that represent actual ground distances.

### 3.0 DIRECTION AND BEARING

- **Cardinal Points:** North (N), South (S), East (E), West (W).
- **Intercardinal (Ordinal) Points:** Northeast (NE), Northwest (NW), Southeast (SE), Southwest (SW).
- **Bearing:** A more precise way of stating direction, measured in degrees (°) from North (0°), moving clockwise. True North is 0°, East is 90°, South is 180°, and West is 270°.

## 4.0 CONVENTIONAL SIGNS AND SYMBOLS

Maps use standardized symbols to represent physical and human features.

- **Point Symbols:** For specific locations (e.g., a church, a school).
- **Line Symbols:** For linear features (e.g., roads, rivers, railways).
- **Area Symbols:** For larger features (e.g., forests, lakes, urban areas).

### COMMON TOPOGRAPHIC MAP CONVENTIONAL SIGNS

| FEATURE               |             | SYMBOL                       |
|-----------------------|-------------|------------------------------|
| Paved Road            |             |                              |
| Unpaved Road          |             |                              |
| Footpath/Trail        |             |                              |
| Perennial River       |             | Intermittent Stream          |
| Perennial River       |             |                              |
| Railroad              |             | Building                     |
| Church                |             |                              |
| Mosque                |             |                              |
| Forest (Deciduous)    |             |                              |
| Swamp/Marsh           | Swamp/Marsh | Contour Lines (Intermediate) |
| Contour Lines (Index) |             |                              |
| Spot Elevation        |             |                              |

**Illustration: Common Conventional Signs** (A table or diagram showing common topographic map symbols for features like roads, rivers, buildings, vegetation, and contours.)

## 5.0 CONTOURS AND RELIEF

- **Contour Lines:** Brown lines on a map that join points of equal height above sea level.
- **Characteristics of Contours:**
  - They never cross.

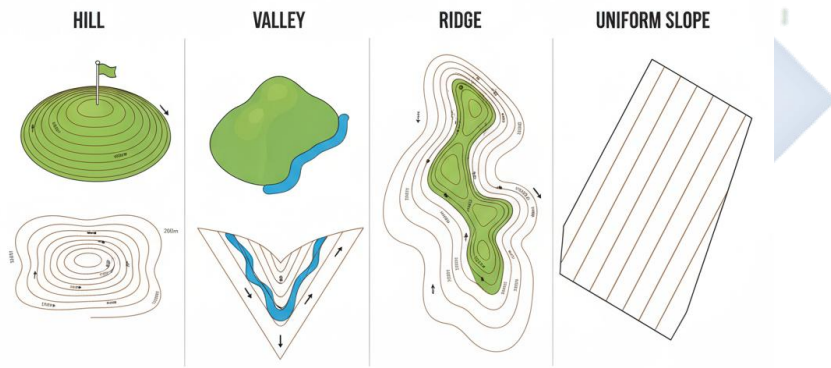
- The closer the contours, the steeper the slope.
- The farther apart the contours, the gentler the slope.

- **Landforms from Contours:**

- **Hill:** Shown by concentric closed contours. The inside has higher values.
- **Valley:** Contours form a V or U shape that points upstream (towards higher ground).
- **Ridge:** A long, narrow hill. Contours are elongated.
- **Depression:** Shown by closed contours with hachures (short lines) pointing inward.



#### CONTOUR PATTERNS FOR COMMON LANDFORMS



**Illustration: Contour Patterns for Common Landforms** (*A diagram showing how a hill, valley, ridge, and uniform slope would be represented by contour lines.*)

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#### EVALUATION (WEEK 3)

1. List four essential components of a map and state the function of each.

2. Convert the statement scale "1 cm to 2 km" into a Representative Fraction (R.F.).
3. What is the bearing of a place that is directly Southeast?
4. How would you distinguish a steep slope from a gentle slope on a topographic map?

***ASSIGNMENT (WEEK 3)***

1. Draw a simple map of your school compound. Include a title, a key with at least 5 symbols, an arrow pointing north, and a linear scale.
2. Practice taking bearings. In your school compound, use a compass to find the bearing of three different landmarks (e.g., the gate, the library, the football goal post) from a central point.



## WEEK 4: MAP DISTANCES

### 1.0 MEASURING STRAIGHT-LINE DISTANCES

This is the shortest distance between two points.

- **Method:** Use a ruler to measure the distance between the two points on the map in centimeters or inches. Then convert this map distance to ground distance using the map scale.

#### Example Calculation:

- Map distance between two towns = 5 cm.
- Map scale: 1:50,000
- Ground distance = Map distance  $\times$  Scale denominator
- = 5 cm  $\times$  50,000 = 250,000 cm
- Convert to kilometers (1 km = 100,000 cm): 250,000  $\div$  100,000 = **2.5 km**

### 2.0 MEASURING CURVED OR WINDING DISTANCES

This is used for roads, rivers, or railways.

- **Method 1 (Using a Thread):**
  - a. Place a thin thread along the route, carefully following all the curves.
  - b. Mark the start and end points on the thread.
  - c. Stretch the thread straight and measure the length between the marks with a ruler.
  - d. Convert this length to ground distance using the scale.
- **Method 2 (Using a Pair of Dividers):**
  - a. "Walk" the dividers along the route, using small, equal steps.
  - b. Count the number of steps.
  - c. Measure the length of one step with a ruler.
  - d. Total map distance = Number of steps  $\times$  Length of one step.
  - e. Convert to ground distance.

### 3.0 CALCULATING AREAS ON A MAP

- **For Regular Shapes (Squares, Rectangles):** Divide the area into regular shapes. Calculate the area of each shape (Length  $\times$  Width) and sum them up. Remember to convert map areas to ground areas using the scale squared.
    - **Ground Area = Map Area  $\times$  (Scale Denominator)<sup>2</sup>**
  - **For Irregular Shapes:** Use the grid square method.
    - a. Place a transparent grid sheet over the area.
    - b. Count the number of full squares inside the area.
    - c. Estimate the number of partial squares (e.g., two half squares = one full square).
    - d. Total Map Area = Total number of squares  $\times$  Area of one square.
    - e. Convert to ground area.
- 

### EVALUATION (WEEK 4)

1. The distance between two villages on a map of scale 1:100,000 is 8 cm. What is the actual ground distance in kilometers?
2. Describe how you would measure the length of a winding river on a topographic map.
3. On a map with a scale of 1:50,000, a rectangular farm measures 4 cm by 3 cm. Calculate the actual area of the farm in square kilometers.

### ASSIGNMENT (WEEK 4)

1. **Practical Exercise:** Use a topographic map of your choice (or one provided by your teacher). a) Measure the straight-line distance between two points. b) Measure the distance along a winding road between the same two points. c) Compare the two distances and explain why they are different.
  2. Calculate the ground area of your school compound using the grid square method on a map.
-

## WEEK 6: MAP REDUCTION AND ENLARGEMENT

### 1.0 CONCEPTS OF MAP REDUCTION AND ENLARGEMENT

- **Map Reduction:** The process of decreasing the size of a map while maintaining the same level of detail, resulting in a smaller representation of the same area.
- **Map Enlargement:** The process of increasing the size of a map to show more detail of a smaller area.
- **Scale Change:** Both processes involve changing the map's scale. Reduction decreases the scale (e.g., from 1:25,000 to 1:50,000), while enlargement increases the scale (e.g., from 1:100,000 to 1:50,000).

### 2.0 METHODS OF MAP REDUCTION AND ENLARGEMENT

**2.1 The Square Grid Method (Graphical Method)** This is the most accurate manual method.

- **Procedure for Enlargement:**
  - a. Draw a grid of equal squares over the original map.
  - b. On a new sheet, draw a grid with the same number of squares but larger dimensions.
  - c. Carefully copy the features from each square of the original map to the corresponding larger square on the new sheet.
  - d. Add all necessary map elements (title, scale, legend, north arrow).
- **Procedure for Reduction:**
  - a. Follow the same steps, but draw the new grid with smaller squares than the original.

**2.2 Mathematical Method** This method uses scale ratios and calculations.

- **Scale Factor:** The ratio between the new scale and the original scale.
  - **For Reduction:**  $\text{Scale Factor} = \text{Original Scale} / \text{New Scale}$
  - **For Enlargement:**  $\text{Scale Factor} = \text{New Scale} / \text{Original Scale}$



- **Example Calculation:**

- Original map scale: 1:50,000
- Desired new scale: 1:25,000 (Enlargement)
- Scale Factor =  $(1/25,000) \div (1/50,000) = 50,000/25,000 = 2$
- This means the map should be enlarged to 2 times its original size.

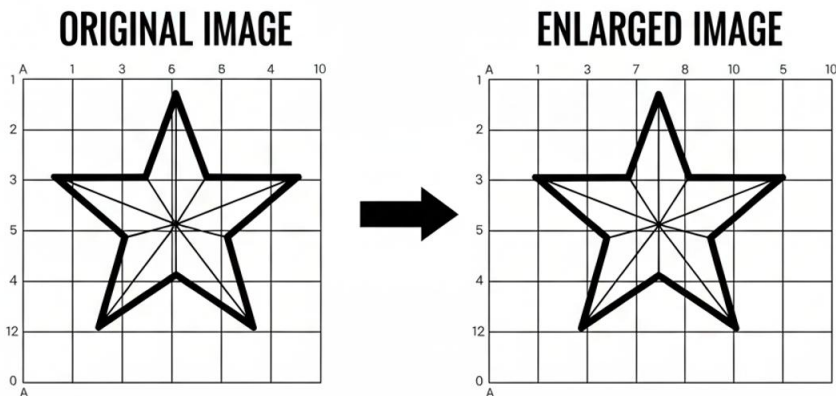
### 2.3 Photocopier Method The quickest modern method:

- Set the photocopier to the desired percentage:
  - For reduction: < 100%
  - For enlargement: > 100%

### 3.0 PRACTICAL APPLICATIONS AND IMPORTANCE

1. **Cartography:** Creating maps at different scales for various purposes (e.g., world maps, country maps, city maps).
2. **Urban Planning:** Enlarging maps to show detailed layouts of specific areas.
3. **Education:** Reducing complex maps to fit in textbooks or worksheets.
4. **Fieldwork:** Creating enlarged maps of specific study areas for detailed annotation.

### Grid Method for for ENLARGEMENT



**Illustration: Grid Method for Enlargement** (*A diagram showing a small grid with a simple feature, and next to it, a larger grid with the same feature drawn proportionally larger.*)

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### ***EVALUATION (WEEK 6)***

1. Define map reduction and map enlargement.
2. Describe the steps involved in enlarging a map using the square grid method.
3. A map with a scale of 1:100,000 needs to be reduced to a scale of 1:200,000. What is the scale factor, and what percentage reduction would you use on a photocopier?
4. Why would a town planner need to use map enlargement?

### ***ASSIGNMENT (WEEK 6)***

1. **Practical Exercise:** Use the square grid method to enlarge a simple map (e.g., of your classroom) by a factor of 2. Use graph paper for accuracy.
  2. Calculate the scale factor required to reduce a map from 1:75,000 to 1:150,000.
-

## WEEK 8: INTERPRETATION OF PHYSICAL AND CULTURAL FEATURES

### *1.0 IDENTIFYING PHYSICAL FEATURES ON TOPOGRAPHIC MAPS*

Physical features are natural landforms and water bodies.

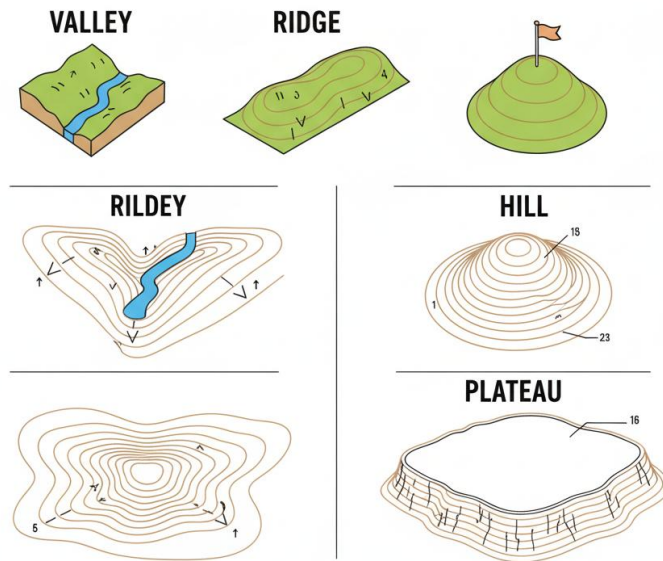
#### **1.1 Relief (Landforms) using Contours:**

- **Hill/Mountain:** Concentric circles of contours with increasing values toward the center.
- **Ridge:** A long, narrow hill with contours elongated in shape.
- **Valley:** Contours form a 'V' or 'U' shape. The 'V' points upstream (toward higher ground).
- **Spur:** A projecting ridge of land. Contours form a 'V' pointing downward.
- **Cliff/Steep Slope:** Very closely spaced contours.
- **Gentle Slope:** Widely spaced contours.
- **Plateau:** An elevated area with a relatively flat top (contours are wide apart) and steep sides (contours are close together).

#### **1.2 Drainage (Water Features):**

- **Rivers and Streams:** Blue lines. Width indicates size.
- **Lakes and Ponds:** Blue enclosed areas.
- **Waterfalls:** Shown where a river crosses contours very closely together.
- **Swamps/Marshes:** Blue areas with marsh symbols (clusters of reeds or grass symbols).

## CONTOUR PATTERNS FOR LANDFORMS



**Illustration: Contour Patterns for Landforms** (A series of small 3D diagrams of a valley, ridge, hill, and plateau alongside their corresponding contour patterns.)

## 2.0 IDENTIFYING CULTURAL (HUMAN-MADE) FEATURES

Cultural features represent human activities and settlements.

- **Settlements:**
  - **Nucleated:** Buildings clustered together (common in rural areas).
  - **Dispersed:** Buildings spread out (common in farmlands).
  - **Linear:** Buildings arranged along a line (e.g., a road or river).
- **Transportation:**
  - **Roads:** Black or red lines (thickness indicates importance).
  - **Railways:** Black lines with cross ties.
  - **Bridges:** A symbol across a river.

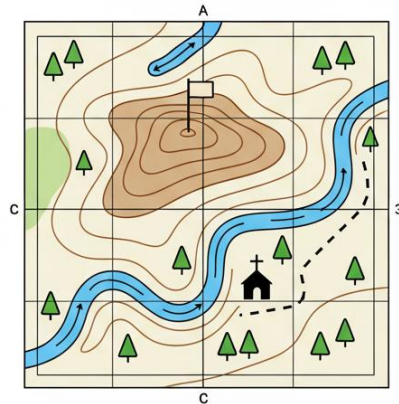
- **Airports:** A symbol of an airplane.
- **Economic Features:**
  - **Quarries/Mines:** A small black pit symbol.
  - **Cultivated Land:** Green areas with small tree symbols or regular patterns.
  - **Factories/Industries:** Small black blocks or special symbols.

### ***3.0 DEDUCING HUMAN ACTIVITIES FROM MAP EVIDENCE***

A key skill is to interpret *why* human features are located where they are, based on the physical geography.

- **Settlement Patterns:**
  - Villages along a river (water supply, transport).
  - Towns at road junctions (trade and accessibility).
  - Settlements on gentle slopes rather than steep slopes (easier to build).
- **Economic Activities:**
  - Quarries on hills (source of rock).
  - Cultivated land on flat plains (easier to farm).
  - Forest reserves in mountainous areas (less suitable for farming).

## MAP EXTRACT ANALYSIS



### QUESTIONS:

1. Identify two physical features shown on map.

Answer: \_\_\_\_\_

2. Identify two cultural features.

Answer: \_\_\_\_\_

Answer: \_\_\_\_\_

3. What is the approximate elevation of the hill at the top of the map? \_\_\_\_\_

4. Describe the relationship between the road and river? \_\_\_\_\_

5. In which grid square is the church located?

**Illustration: Map Extract Analysis** (A small section of a topographic map with questions guiding students to identify physical features, cultural features, and deduce the relationship between them.)

### EVALUATION (WEEK 8)

1. How would you distinguish between a valley and a ridge on a topographic map?
2. List three cultural features you might find on a topographic map and the symbols used for them.
3. Looking at a map, you see a village located at a river confluence (where two rivers meet). Suggest two reasons why this is a good location for a settlement.
4. What does the presence of a quarry on a hillside and a railway line running nearby tell you about the economic activities in that area?

### ***ASSIGNMENT (WEEK 8)***

1. **Map Interpretation Exercise:** Using a provided topographic map (e.g., a sample WAEC map extract): a) Identify and list three physical features. b) Identify and list three cultural features. c) In a short paragraph, describe the main human activities you think take place in the area shown on the map, justifying your answer with evidence from the map.
  2. Draw a simple sketch map of an imaginary area that includes a river, a hill, a main road, a nucleated settlement, and an area of cultivated land.
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## WEEK 9: TRANSPORTATION

### 1.0 DEFINITION AND MODES OF TRANSPORTATION

**Transportation** is the movement of people, goods, and services from one location to another. It is a vital component of a country's infrastructure and economic development.

#### The Four Main Modes of Transport:

##### 1. Land Transport:

- **Road Transport:** Vehicles using roads (cars, buses, trucks). The most flexible and common mode.
- **Rail Transport:** Trains on fixed tracks. Efficient for heavy goods and long distances.

##### 2. Water Transport:

- **Inland Waterways:** Boats and canoes on rivers and lakes.
- **Ocean/Sea Transport:** Ships for international trade.

##### 3. Air Transport: Airplanes and helicopters. The fastest mode, ideal for perishable goods and passengers.

##### 4. Pipeline Transport: Underground pipes for transporting liquids and gases (crude oil, natural gas, water).

### 2.0 ADVANTAGES AND DISADVANTAGES OF EACH MODE

#### Comparative Table of Transport Modes:

| Mode         | Advantages                                                                         | Disadvantages                                                                 |
|--------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Road</b>  | Flexible, door-to-door, widely available.                                          | Congestion, high accident rate, affected by weather, high maintenance cost.   |
| <b>Rail</b>  | High capacity for bulky goods, relatively safe, cost-effective for long distances. | Inflexible (fixed routes), high initial cost, slow for short distances.       |
| <b>Water</b> | Cheapest for bulky goods, high capacity, low energy consumption.                   | Very slow, limited to routes with water, requires ports, affected by weather. |



| Mode            | Advantages                                                                 | Disadvantages                                                        |
|-----------------|----------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Air</b>      | Fastest, good for perishables and high-value goods, unaffected by terrain. | Most expensive, limited capacity, requires airports, high pollution. |
| <b>Pipeline</b> | Low operating cost, continuous flow, safe for hazardous materials.         | High initial cost, inflexible, can be vulnerable to vandalism/leaks. |

### ***3.0 FACTORS INFLUENCING THE DEVELOPMENT OF TRANSPORT NETWORKS***

#### **1. Physical Factors:**

- **Relief:** Flat land is easier and cheaper to build on than mountainous terrain.
- **Climate:** Heavy rainfall can damage roads and disrupt transport.
- **Rivers and Seas:** Provide natural routes for water transport but can be obstacles for land transport, requiring bridges.

#### **2. Economic Factors:**

- **Availability of Capital:** Building transport infrastructure is very expensive.
- **Volume of Trade:** Areas with high economic activity need better transport links.
- **Presence of Resources:** Resources like oil and minerals often necessitate the development of pipelines, railways, or ports.

#### **3. Political and Social Factors:**

- **Government Policy:** Investment in infrastructure is a key government decision.
- **Population Distribution:** Densely populated areas require denser transport networks.

### ***4.0 TRANSPORTATION IN NIGERIA: PROBLEMS AND SOLUTIONS***

#### **• Problems:**

- a. **Poor Road Conditions:** Many roads are in disrepair, leading to accidents and high vehicle maintenance costs.
- b. **Inadequate Rail Network:** The rail system is underdeveloped and outdated.
- c. **Port Congestion:** Major seaports (Lagos, Port Harcourt) experience delays.
- d. **Insecurity:** Highway robberies and kidnappings on some routes.

#### **• Solutions:**

- a. **Increased Funding and Maintenance:** Regular maintenance of roads and railways.
  - b. **Modernization:** Electrification and expansion of the railway network.
  - c. **Private Sector Participation:** Public-Private Partnerships (PPPs) to fund and manage infrastructure.
  - d. **Improved Security:** Increased patrols on major transport routes.
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### ***EVALUATION (WEEK 9)***

1. List the four main modes of transportation.
2. Compare road transport and rail transport, giving two advantages of each.
3. Explain how physical factors like relief and climate can influence the development of transport in a region.
4. Identify two major problems facing transportation in Nigeria and suggest a solution for each.

### ***ASSIGNMENT (WEEK 9)***

1. Create a poster or a short presentation advocating for the improvement of one mode of transport in Nigeria (e.g., "Why Nigeria Needs a Modern Railway System"). Include the benefits and potential challenges.
  2. **Case Study:** Research a major transport project in Nigeria (e.g., the Lagos-Ibadan Railway, the Second Niger Bridge). Write a one-page report on its objectives, current status, and potential impact on the economy and people.
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